Cost Estimating Manual for Projects

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Development Division

Foreword

WSDOT's vision is to be the best at providing a sustainable and integrated multimodal transportation system. Its mission is to provide and support safe, reliable, and cost-effective transportation options to improve livable communities and economic vitality for people and businesses.

Estimators should be shielded from pressures to keep estimates within desired amounts.

The WSDOT Cost Estimating Manual for Projects is updated periodically as part of an ongoing process to keep technical information relevant and current, as well as provide consistency with the department's vision.

More of an art than a science, cost estimating requires a thorough understanding of project scope, past price history, and current market conditions, as well as a generous application of human judgment.

Key goals of the estimating process described in this manual include:

1. Department-wide priority on estimating, managing, and controlling costs:
    - Fully developed and integrated policies, processes, and tools for cost estimation, management, and control
    - Baseline estimates that align with early project scope development and include an initial assessment of uncertainty
    - Clearly defined and documented cost management endorsement processes to authorize changes in scope and cost after the baseline estimate is established
    - Dedicated resources that are focused on effective scoping, project cost estimating, and cost management
2. Reliable and accurate estimates:
    - Documented and complete cost estimates that include costs for Preliminary Engineering (PE), Right of way (RW), and Construction (CN).
    - Clearly spelled-out assumptions, risks, and uncertainties that can be easily communicated
3. Statewide uniformity and consistency:
    - Uniform application and consistent statewide use of well-documented processes, tools, and templates
    - Use of processes and tools during the planning, scoping, design, and letting phases
4. Improved communication and credibility with external stakeholders:
    - Consistent and clear communication of cost estimate range to external stakeholders at milestone points
    - The ability to communicate cost estimates with confidence, leading to stronger relationships with external stakeholders, to encourage greater possibility for collaboration, and increased funding support of transportation initiatives
5. Clear accountability:
    - Accountability for cost estimating and cost management at all levels of the agency
    - Defined roles and responsibilities for every person involved
    - Accountability that is tracked at key milestones in the process

This manual is intended as a guide for developing and managing project cost estimates. It provides policies, rules, procedures, and tools to aid WSDOT staff in preparing project cost estimates and managing cost throughout WSDOT's project development process. Although the manual is primarily written for project managers and estimators, it is strongly recommended that others involved in WSDOT's project delivery activities familiarize themselves with this information.

The term "project cost estimate," as used during the project development process, includes all direct capital outlay costs, including right of way, structures, and landscaping, but does not normally include indirect capital outlay support costs. Indirect Capital Support costs are business or operational expenses that are incurred for a common or joint purpose and are **not readily identifiable** or directly connected with a single, specific project. These costs, often referred to as overhead or administrative costs, are necessary for the general operation of the organization and the performance of its programs as a whole.

Project cost estimates should be as accurate as possible. Estimates should never be artificially reduced to stay within the funding limits, nor should they be reduced to make available more funding for region projects. Likewise, project cost estimates should not be artificially raised beyond the contingency percentages provided for in this manual unless adequately justified.

Important Notice

The detailed estimate of any phase of an estimate shall not be disclosed to the public and will be kept confidential until the end of the bid opening.

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Chapter 1 **Purpose**

Cost estimating is the predictive process used to quantify cost, and price the resources required to develop and deliver projects.

The need to solidify the estimation process can be seen in four areas:

1. Financial plan – cost estimates are used to obtain and allocate funding for projects.
2. Public trust – relies in large part on our ability to deliver transportation projects in a timely manner and within budget. Declarations of project cost set public expectations. Estimate ranges need to be thoughtfully provided using a well-documented, quality estimate.
3. Project controls – rely on cost estimates to help keep projects within the appropriate fiscal boundaries. The baseline project cost estimate helps keep the project from growing and expanding beyond spending limits.
4. Project resilience – as projects encounter problems, estimates are scrutinized. The quality of the estimate and the documentation provide critical support to project success.

The reliability of project cost estimates at every stage in the project development process is necessary for responsible fiscal management. Unreliable cost estimates result in severe problems in programming and budgeting. Estimates affect staffing and budget decisions that could impair effective use of department resources. Communication with decision makers on critical budget issues is also adversely affected.

This document offers guidance and instruction for design staff and project managers — a consistent approach to cost estimating, including:

- How to treat common and recurring challenges encountered in the estimating process.
- Considerations, tips, and cautions for different estimate phases.
- Resources.

Note: Guidance for calculating quantities of a proposed design is found in the WSDOT Design Manual.

Chapter 2 Cost Estimating Process

2.01 Introduction
2.02 Gather Project Information
2.03 Document the Basis of Estimate
2.04 Prepare Cost and Schedule Estimate
2.05 Review Cost and Schedule Estimate
2.06 Perform Risk Assessment
2.07 Determine Communication Approach
2.08 Obtain Management Endorsement

Exhibit 2-1 Estimating Process
Exhibit 2-2 Components of Project Cost Estimates

The most common question asked of any project at any stage of design is "How much is it going to cost?" followed by "How long is it going to take?" and finally "Why?" The next few pages describe how to prepare an estimate so these questions can be answered confidently. Good estimating requires a professional and deliberate approach. Information is at the heart of the cost estimating process.

The cost estimating process can be organized into 7 steps:

1	2	3	4	5	6	7
Gather Project Information	Determine the Basis of Estimate	Prepare Cost and Schedule Estimate	Review Cost and Schedule Estimate	Perform Risk Assessment	Determine Communication Approach	Obtain Management Endorsement

2.01 Introduction

Take the necessary steps to gather project information and organize to estimate project cost. Exhibit 2-1 illustrates WSDOT's cost estimating process. This process is applied to all phases of project development, from planning through Plans, Specifications, and Estimates (PS&E).

Each project phase, Preliminary Engineering (PE), Right of way (RW), and Construction (CN) may require different estimating inputs, methods, techniques, and tools. The task of cost estimating, by its very nature, requires the application of prudent judgment. Following is a short description of each step in the cost estimating process.

2.02 Gather Project Information

Make estimating a priority and utilize appropriate time and resources. A large amount of project information is typically available from various sources, including:

- Project Management Guide
- Basis of Design (BOD)
- Context and Modal Accommodation Report
- Intersection Control Evaluation
- Risk Register
- Project Plans
- Project specific reports, bullet lists
- Agreements
- Risk Breakdown Structure (RBS)
- Project Charter

- Master Deliverables List (MDL)
- Stakeholder Register
- Design Documentation Package
- Subject Matter Experts
- Contractor view of cost (or someone with significant construction experience)
- A variety of other reports or studies.

The information should describe the project need, schedule, scope, and context. Review the cost estimating process and outline a reasonable schedule to prepare the cost estimate commensurate with the level of project development. Consider the following questions while getting organized for the cost estimating effort: Why do we need information? Where does the information come from? What type of information is needed? What subject matter experts are available to assist with the information (for example, HQ Railroad Liaison…)? Review and transform the information into an easily used and communicated product.

2.03 Document the Basis of Estimate

The Basis Of Estimate (BOE)documentation enables the agency to easily track changes to project scope, cost, and schedule. This document provides a history of what is known about the project, and documents its evolution. Project "knowns" as well as "unknowns" are clearly identified. It backs up the assumptions, quantities and costs shown in the estimate. It answers the questions of why the project costs what it does and the time it takes to deliver. This record is important because multiple estimators may be involved on the project, and complex projects can take years to develop. The BOE and estimate should be periodically updated to stay abreast with the project's development.

2.04 Prepare Cost and Schedule Estimate

Now that we have readied ourselves via the basis of estimate and assembled project information, preparation of the project cost estimate can be conducted in earnest. To the extent known, based on the project scope and limits, bid items can be identified and quantified for estimating, applying standard bid items, using quantities and unit prices, or for earlier cost estimating efforts parametric estimates can be prepared. Each bid item in the estimate should have a calculation sheet that includes information about the quantities and costs used. For example, explain how quantities and costs were determined and any specifics used to make a judgment about costs, including approach to rounding.

For some projects an overall cost estimate is prepared and then percentages are used. See the table in Exhibit 7-3 for a markup summary. Items that can be calculated using percentages include Design Allowance, Preliminary Engineering, Mobilization, Sales Tax, Contingency, and Construction Engineering. When preparing your cost estimate, recognize that you are not alone. If you are estimating a project type that you do not have experience with, be sure to ask colleagues, supervisors, and pertinent specialty group experts about the project and items that need to be included.

The cost estimate is determined in current year dollars and is often prepared using a variety of different techniques—depending on the scope definition, project size, and complexity of the project. As the design progresses and more details are known, project components become more detailed. Inputs include project scope details, historical databases, and other cost databases (such as bidtabs.net, WSDOT Unit Bid Analysis and

RS Means Cost Data). WSDOT specialty groups, relevant to the project, should be consulted when preparing base cost estimates.

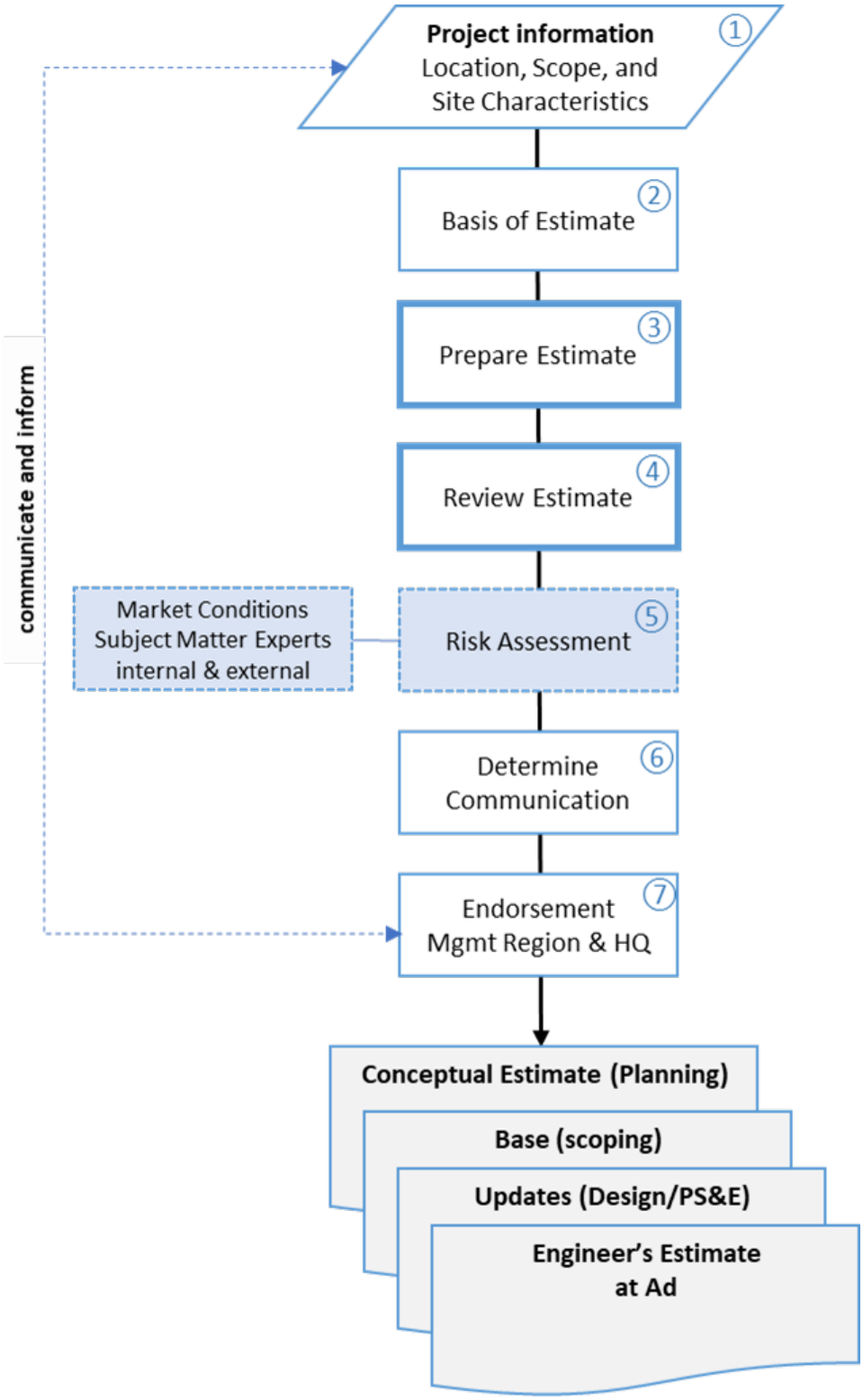
The estimate is also dependent on the project schedule. A high-level summary of the estimated schedule should be attached to the Basis of Estimate. At a minimum, schedules developed in the early stages of project development should include the major milestones that WSDOT uses to measure performance and progress on projects. An early schedule should include major milestones but may only include a few activities equal with the level of development the project has reached. Typically, at a minimum, an early project schedule should include estimated durations for the environmental, design, ROW, ad/bid/award, and construction phases.

Per the Program Management Manual, PMM, Chapter 10, section 10-5.3 there are six reportable milestones for line-item projects, indicated via bold text and asterisk in the milestone list below.

Project Definition Complete*	**Right of Way Certification***
Begin Preliminary Engineering* (PE)	**Advertisement Date***
Environmental Documentation Complete*	**Award**
Right of Way (RW) Plan Approved	**Right of Way (RW) Phase End**
Right of Way Phase Start	**Construction Work Start**
Preliminary Engineering (PE) Phase End	**Operationally Complete Date***
Environmental Permits Received	**Construction (CN) Phase End**

Exhibit 2-1 provides a flowchart of the Cost Estimating Process.

Exhibit 2-1 Estimating Process



2.05 Review Cost and Schedule Estimate

This activity confirms that:

1. Project and estimate assumptions, as documented in the Basis of Estimate, are appropriate for the project.
2. The base cost estimate is an accurate reflection of the project's scope of work.
3. Scope, schedule, and cost items are calculated properly, and required components are not missing or double counted
4. Historical and other data used is realistic in applicability of costing time frame and quantities, and that it reflects project scope and site conditions. Internal specialty groups and/or subject matter experts (SMEs) must participate in the base estimate review.

The estimate review plays an important role in ensuring the quality of estimate and contributes to the overall quality of the PS&E package.

2.06 Perform Risk Assessment

This activity uses a structured approach to manage project cost and schedule uncertainties. WSDOT policies define the timing and methodology for risk assessments, offering guidance on budget adjustments to accommodate uncertainty and risk planning. The WSDOT Project Risk Management Guide provides detailed instructions for applying risk assessment methods effectively.

2.07 Determine Communication Approach

In general, the following guidance is provided when making public declarations of cost estimates:

1. An estimate is a range, not a single number.
2. An estimate for a project should include everything: PE, RW, and CN.
3. An estimate should be communicated in Year of Expenditure dollars.

Regardless of whether qualitative or quantitative risk assessment is employed, project cost estimate data normally needs to be communicated to both internal and external stakeholders.

The communication approach determines what estimate information should be shared, who should receive this information, and how and when the information should be communicated.

Cost estimate information should be included when the communication plan is developed as part of the project management process. Often the words are as important as the numbers. The Basis of Estimate document is useful in documenting and monitoring key information about the project.

2.08 Obtain Management Endorsement

Estimates are key products of the project management process and are fundamental documents upon which management decisions are based. Given their importance, all estimates should be reviewed by an independent party, then be reconciled and revised as needed to respond to independent reviewers' comments. Once the independent review comments have been addressed, estimates should be presented to management staff for their endorsement.

Management endorsement of estimates developed for initial budgeting or base definition is defined as a step in the project management process. Revised estimates typically developed if project requirements change, or as design is developed, should also be reviewed by management staff, revised as necessary to reflect management

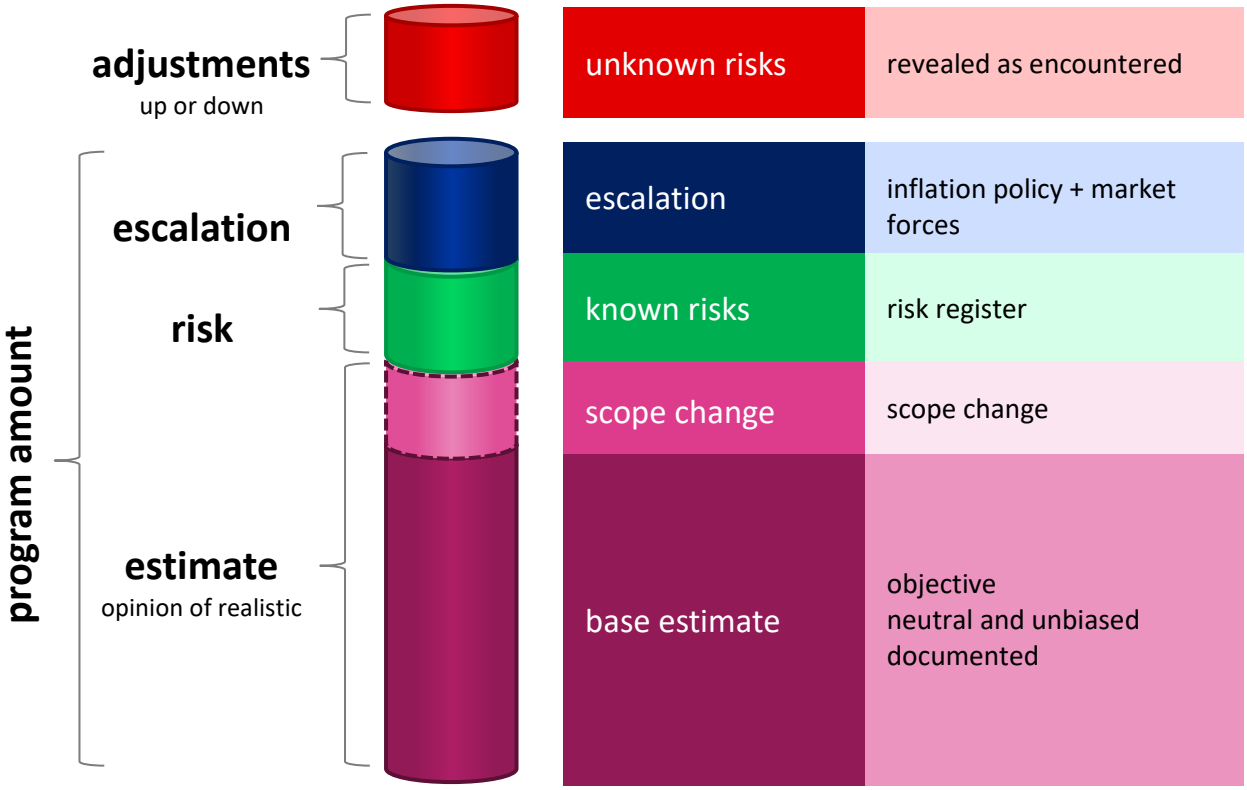
comments, and endorsed. Revised estimates are incorporated into project cost estimates through the project change management process.

Estimates are not altered to fit a predetermined budget outside of the formal estimating process. In the event that management endorsement of an estimate cannot be obtained, revisit the estimating process starting with the project definition step (project information, project scope, project location, and site characteristics) in Exhibit 2-1.

Communication and management endorsement are closely linked. There are at least two aspects of communication that are very important to consider. The first is internal communication, this involves communication of the project costs among the project team, the project manager and also the specialty groups that may have helped in developing the estimate. In many cases the project cost estimate is a compilation of estimates from a number of sources. Secondly the project costs are communicated publicly. This should be done only after an endorsed communication plan has been prepared. It is our duty to communicate project elements such as cost and schedule in a clear and coherent matter and to ensure the information provided is accurate and of high quality.

Management may be interested in better understanding the estimating and how the opinion of cost was developed. In addition, management may desire to know the components of the estimate and how the final figures are developed.Exhibit 2-2 offers an at-glance view of the components of a project cost estimate.

Exhibit 2-2 Components of Project Cost Estimates



Chapter 3 Cost Estimating Methodology

3.01 Introduction
3.02 Parametric
3.03 Historical Bid-Based
3.04 Cost-Based
3.05 Risk-Based
3.06 Specialty Items
3.07 Three-Point Estimating

Exhibit 3-1 Three Point Estimating spreadsheet

3.01 Introduction

Estimating methodologies are applied to individual bid items, and may change as the project develops:

1. parametric
2. historical bid-based
3. cost based
4. risk based

Each method offers techniques and tools to aid the estimator. A combination of methods may be found in any given estimate.

3.02 Parametric

Parametric methods are applied to projects in the planning, scoping, or early design stage. These methods involve techniques that use historical data to define the cost of the typical transportation facility using measurements that are easily determined, such as cost per lane mile, cost per interchange, cost per square foot, and cost per intersection.

Two techniques are commonly used in parametric estimating:

1. analogous (similar) projects, and
2. historical percentages.

WSDOT uses an Access database estimating program to support parametric estimating: Planning-Level Project Cost Estimating (PLCE)

This tool is only available to WSDOT personnel on the following site:

Planning-Level Project Cost Estimating (PLCE)

3.03 Historical Bid-Based

Historical bid-based methods are commonly used to develop WSDOT cost estimates and are appropriate when design definition has advanced to the point where detailed quantification of bid items is possible. The unit cost prices used are collected and stored from prior projects. They should be modified or adjusted to reflect current prices and project-specific conditions such as geographic location, quantity of item(s) needed, and the scheduled timing of project advertisement. Techniques such as historical bid pricing, historical percentage, and cost-based estimating are also used to determine unit prices.

Estimating data sources include:

- Oman Systems BidTabs.Net – Bid History
- WSDOT Unit Bid Analysis (UBA) – Bid History
- RS Means Cost Data – Cost Based, when WSDOT-specific unit costs are not available (this tool can be used for both historic bid-based and cost-based methods)
- Risk Modeling – WSDOT Project Risk Analysis Model, PRAM tool, risk-based project cost estimating

3.04 Cost-Based

Cost-based estimating methods involve calculating the contractor's expenses for materials, equipment, and labor for a specific item or group of items. To these costs, an estimated amount for contractor overhead and profit is added. This approach is particularly useful in cases where historical unit price data is unavailable or not suitable for the project at hand. Contractors often rely on cost-based estimating to develop their bids. It also serves as a tool to guide decisions on contract awards or rejections and aids in potential price negotiations with the contractor after the contract is awarded.

Typically, cost-based estimates focus on the highest-value items in a project, those that make up approximately 20% of the items but account for 80% of the total project cost. For the remaining items, historical bid-based estimating methods can be applied, optimizing the use of resources and minimizing the time and cost involved in preparing cost-based estimates. Additionally, cost-based estimating provides a means to validate key items in a historical bid-based estimate, ensuring the continued relevance and accuracy of historical pricing data.

3.05 Risk-Based

Risk-based estimating methods use the probabilistic relationships between base cost, base durations, and risks related to the project. This approach may incorporate a variety of techniques, including historical data, cost-based estimating, and the best judgment of subject matter experts for given types of work, to develop the base cost (the cost of the project if all goes as planned). Risks (opportunities or threats) are then defined and applied to the base cost through modeling (Monte Carlo simulation) to provide a probable range for both project cost and schedule. The schedule and cost are integrated into the same model.

For more information about risk-based estimates, please see the links provided in Section 2.06.

3.06 Specialty Items

There are a number of specialty items in WSDOT projects that are estimated by designated offices. These items include bridges and non-standard retaining walls, traffic items, right of way and other properties, environmental features, and utilities. The procedures to estimate these items are covered in Chapter 8 – Specialty Groups.

3.07 Three-Point Estimating

Three-point estimating, in this application, focuses on natural variations in unit prices, particularly when analyzing historical bid data. This approach accounts for the inherent variability in pricing due to project specific conditions, supplier differences, or regional factors. Here is how it is used:

Optimistic Estimate (O): This represents the lowest unit price observed in the bid history database. It reflects the most favorable market conditions or supplier offers.

Most Likely Estimate (M): This is the average or median unit price from the database, representing the typical market scenario.

Pessimistic Estimate (P): This is the highest unit price recorded, accounting for less favorable conditions or higher-cost suppliers.

Three Point Estimating Formula: E = (O + 4M + P) / 6

Using this method on historical data delves into the natural variations found within the unit bid history database and synthesizes these data points into a unit price for the estimate being developed.

- **Understanding the Data:** This method analyzes historical unit prices to account for variability caused by project conditions and regional influences, aiming to determine a representative price.
- **Synthesizing the Data**: This approach distills the historical data into a unit price informed by, and taking into account the natural variation. This synthesized price might be a weighted average, based on specific criteria (e.g., recent bids, project location, and quantity).
- **Practical Application**: This synthesized unit price becomes a practical input for cost estimation. This can be a valuable tool in developing a project base cost estimate that may be used in a Cost Risk Assessment of Cost Estimate Validation Process.

This approach complements quantitative risk assessments and contributes to quality base cost estimates free from biases. This neutrality ensures that the estimate reflects real-world conditions and minimizes subjectivity. The base estimate serves as a foundational input for risk-based estimating, which is where variability, uncertainty, and potential risks are thoroughly analyzed.

This methodology is particularly beneficial because it separates the neutral data-driven analysis from the subjective elements of risk assessment. It ensures clarity and transparency in the estimation process, and in communicating the estimate.

The three-point estimating technique is particularly effective for Design-Build (DB) highway construction projects for several reasons:

1. **Estimating in Greater Uncertainty**
   Design-Build (DB) project estimates are based on preliminary information and conceptual designs, resulting in greater uncertainty compared to Design-Bid-Build (DBB) projects. Three-point estimating offers a more comprehensive perspective on potential costs.
2. **Risk Management**
   DB projects have overlapping design and construction phases—this can introduce risks. Evaluating costs and timelines as a range more realistically reflects project risk and uncertainty.
3. **Adaptable, Flexible, and Nimble**
   In DB projects, the owner concept design is speculative. The design evolves as DB Teams make their proposals and offer Alternative Technical Concepts. Three-point estimating naturally allows for nimble and responsive adjustments to cost estimates. The cost range produced ensures that estimates remain realistic as new ideas and information emerge.
4. **Improved Decision Making**
   A single-point estimate masks the inherent uncertainty in project costs—three-point estimates help produce more informed decisions by considering the various outcomes for each item in the cost estimate.
5. **Communication**
   Three Point estimating offers more transparency and simplifies the explanation of potential cost ranges to stakeholders.

In summary, the three-point estimating technique enhances the accuracy, flexibility, and reliability of cost estimates for DB highway construction projects.

For the reasons cited above, three-point estimating is recommended for determining the project base cost estimate.

Exhibit 3-1 Three Point Estimating spreadsheet

No.	Std Item Number	Description	UNITS	Estimated Quantity			Estimated Unit Cost			Estimated Amount			PERT
				Optimistic (minimum)	Most Likely	Pessimistic (maximum)	Optimistic (minimum)	Most Likely	Pessimistic (maximum)	Optimistic (minimum)	Most Likely	Pessimistic (maximum)	Estimated Value
1	0310	ROADWAY EXCAVATION INCL. HAUL	C.Y.	4000	7000	9000	$40.00	$44.00	$54.00	$160,000.00	$308,000.00	$486,000.00	$313,000.00
2	4322	CONC. CLASS 4000 FOR BRIDGE	C.Y.	1000	1600	2000	$1,200.00	$1,300.00	$1,600.00	$1,200,000.00	$2,080,000.00	$3,200,000.00	$2,120,000.00
3	5100	CRUSHED SURFACING BASE COURSE	TON	2500	3000	4000	$45.00	$47.00	$52.00	$112,500.00	$141,000.00	$208,000.00	$147,416.67
4	5625	CEMENT CONC. PAVEMENT	C.Y.	2500	3000	4000	$425.00	$430.00	$490.00	$1,062,500.00	$1,290,000.00	$1,960,000.00	$1,363,750.00
5	5766	HMA CL. 3/8 IN. PG	TON	12000	16000	00	$112.00	$123.00	$140.00	$1,344,000.00	$1,968,000.00	$2,660,000.00	$1,979,333.33
6													
7													
8													
9													

EXAMPLE ONLY

Three Point Estimating

Utilization of three estimates is to define an approximate range for an item's quantities, and costs to improve upon the accuracy of a single-point estimate. This concept originates from the Program Evaluation and Review Technique (PERT).

PERT Beta Distribution Estimation Equation:

$$E = \frac{(O + 4M + P)}{6}$$

Most Likely (M): Most realistic expectations of quantities/costs

Optimistic (O): Best-case scenario of quantities/costs

Pessimistic (P): Worst-case scenario of quantities/costs

Note: Considerations in determining the bid prices include:

- geographic location
- quantity of item
- item availability
- type of qty pricing
- bidding environment

Chapter 4 Project Development Phases and Estimating

4.01 Introduction
4.02 Preconstruction General Guidance
4.03 Planning
4.04 Scoping
4.05 Design
4.06 PS&E

Exhibit 4-1 Gantt Diagram of Preconstruction Phases
Exhibit 4-2 Cost Estimating Matrix

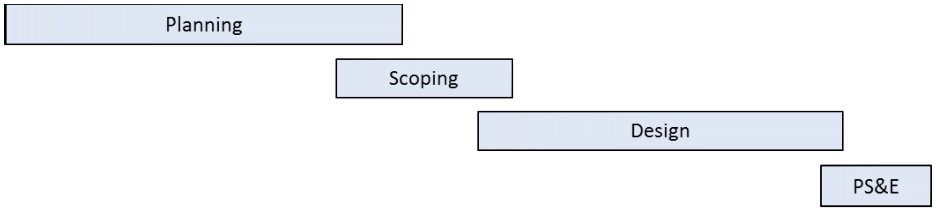
4.01 Introduction

Typical phases of project development in the preconstruction phase are:

1. Planning
2. Scoping
3. Design
4. Plans, Specifications, and Estimates (PS&E)

Exhibit 4-1 illustrates how the preconstruction phases overlap and succeed each other.

Exhibit 4-1 Gantt Diagram of Preconstruction Phases



Blending both art and science, cost estimating demands a comprehensive understanding of project scope, historical pricing data, and current market conditions—combined with the thoughtful application of human judgment. Although the scientific aspects of estimating can be relatively straightforward, the artful elements often require significant time and experience to develop.

4.02 Preconstruction General Guidance

WSDOT provides a number of useful tools and resources, including the documents listed in the following section, which are accessible at: Cost Risk Assessment

4.02(1) WSDOT Estimating and Related Documents

Estimating Documents:

- Project Management Guide
- Project Risk Management Guide
- Cost Estimating Manual for WSDOT Projects
- Basis of Estimate Form
- Estimate Order of Calculations, Tax Rates, & 700 & 800 Items (see appendix A)
- Design Analysis - Estimating Template
- Estimate Checklist
- Design Guidance and Support

4.02(2) Estimating Steps

Regardless of the project development phase of an estimate, a good estimator must consider the following:

- Review the "Estimating Template" (which includes the "Estimate Summary") and "Estimate Order of Calculations, Tax Rates, & 700 & 800 Items" forms, and seek clarification if necessary. Also review the other resources at "Estimating Forms": Cost Risk Assessment
- When determining a project's Basis of Estimate, see the Basis of Estimate form and complete the following:
- Visit the project site in person. See Tips in Section 4.03(2). Regardless of the project development phase, a skilled estimator should consistently apply the following practices:
- Review Estimating Resources
- Consult the Estimating Template, this includes the Basis of Estimate (BOE), Estimate Summary, Order of Calculations, Tax Rates, & 700 & 800 Items forms.
- Seek clarification if necessary.
- Review additional resources available at: Cost Risk Assessment.
- Establish the Basis of Estimate (BOE)
- Conduct a Site Visit – Visit the project site in person. See guidance in Section 4.03(2).
- Review Documentation – Examine all available project definition documents and design parameters.
- Assess Location Characteristics – Consider unique factors such as land use context, travel modes, remoteness, and construction challenges.
- Utilize Visual Resources – Review aerial photographs and/or video logs. Tools such as SRView can be particularly useful.
- Document Assumptions – Record initial estimating assumptions, noting factors that require further investigation.
- Establish an Estimate File – Create and maintain a dedicated file to house estimate calculations throughout the project lifecycle.
- Apply Estimating Methods – Use appropriate estimating techniques to ensure accuracy and consistency.
    - Review all available documents and design parameters relating to project definition.
    - Account for unique project location characteristics, such as land use context, travel modes, remoteness of location and construction challenges.
    - Review aerial photos and/or video logs. SRView can be a useful tool.
    - Document initial estimating assumptions, including factors that will need further investigation.
    - Establish an estimate file in which estimate calculations will be saved over the life of the project.
    - Use the estimate checklist to ensure all essential estimate elements have been reviewed and captured.

When preparing the cost estimate, document the preconstruction phase (see Exhibit 4-1). Document the baseline estimate in each account (PE, ROW, and CN) and the contingency amount separately to facilitate tracking of deviations.

1. When reviewing the total estimate, engage knowledgeable and experienced individuals who are independent from the project. Perform reviews in each project development phase.
2. Use the tools and documents that have been prepared by others to your advantage. Review estimates from similar projects and use as a starting point for items and costs. Use spreadsheets and databases others have developed as templates for building your estimate.
3. Use the expertise of specialty groups, particularly construction, at WSDOT to provide input to your estimate. Special items may have a significant effect on a project's cost. The information obtained from

specialists can greatly increase the accuracy of an estimate. Be sure to inform the specialty groups about any unique issues or uncertainties that you think might affect their estimates.

4. Ask specialty groups about what contingency percentage they recommend for the specialty work and justification for it.
5. Economy of scale plays a large factor in estimating projects. When using a previous project's data to determine unit prices, be sure both projects are similar in size for the given item.
6. Have someone check your work. One of the easiest and most often overlooked ways to eliminate errors is simply to have your estimate reviewed by an experienced co-worker. Have the person look at the spreadsheets as well as your backup calculations.
7. Become familiar with the region in which the project will be constructed and any unique local policies. Market conditions may be different across the state, which will affect the cost of construction and/or materials. Knowing the general locations where materials might be obtained could be beneficial. Even a basic familiarity with contractors and their past practices can help you understand how they may bid on your project.
8. Use checklists throughout the estimating process to make sure no basic items are left out of an estimate.
9. Make sure the appropriate human and technical resources are dedicated to the estimating process.
10. Allow adequate time to estimate the costliest items; in most cases, they have the most important influence on overall cost.
11. Incorporate contractor perspectives into the Engineer's Estimate by consulting appropriately qualified Subject Matter Experts (SMEs) or internal staff with relevant construction and contracting experience. This practice helps validate assumptions, improve estimate accuracy, and ensure alignment with current market conditions.

Cost Management - Regarding cost management, it is wise to consider the following procedures:

1. Obtain endorsements at milestone points during project development. Consider using peer reviews prior to key milestones. If an endorsement is not indicated, the project team may also see value in a peer review involving knowledgeable and experienced individuals from outside the project team.
2. Establish good communication practices within the team and with management about the estimate, including assumptions, uncertainty, and other issues that may influence decisions about the project.
3. Protect estimators from internal and external pressures to deviate from established estimating guidelines, in order to foster and promote objectivity and impartiality.
4. Document updates to the base estimate and the use of miscellaneous items in each project development phase (planning, scoping, preliminary design (30%, 60%, 90%, ad ready), with reference to the scope, cost, and schedule that correspond to the prior endorsed estimate.
5. Contingencies identified in the planning or scoping phases that are not needed at the design or advertising phases should not be used to increase project scope. Return any unused contingency to region program management.
6. Keep region program management and HQ Capital Program Development and Management informed about project changes and external impacts that affect the baseline/project budget. Provide procedures that require endorsements prior to budget changes.

Each project development phase is inherently different from the others; therefore, they all have unique challenges to overcome during the estimating phases. In the following pages, techniques and considerations related to the best estimating practices are presented for each phase. The considerations include: (1) general advice, (2) tips, and (3) cautions.

Each consideration can be used to either increase the quality of the estimate or reduce the time spent on the estimate. The cautions focus on situations that can be detrimental to the quality of an estimate. They generally include warnings about improper use of information, external factors that can affect an estimate, or even common pitfalls the estimators have encountered in the past.

While the list of considerations is a good compilation of pointers and warnings, the best advice will come from an experienced estimator who has worked on similar projects.

4.03 Planning

The planning phase estimate is used to estimate the funding needs for long-range planning and to prioritize needs for the Highway System Plan. These estimates are typically prepared with limited project information.

4.03(1) Techniques

Parametric estimating techniques are commonly used for planning estimates. Lane mile and square foot are two examples of parametric estimating techniques. Historical bid prices and historical percentages can also be used to generate planning level costs estimates. Analogous project estimating, using a similar project to establish costs, is another approach that can be used at the planning phase. Analogous estimating is used if there is limited information about the project

4.03(2) Considerations

General Advice

Analogous projects: The chosen historical project must be truly comparable. Determine the similarities and differences between the historical project(s) and the current project before using historical data.

Older projects: Project data from older projects is less reliable due to variations in prices, standards, inflation, construction technology, and work methods. The analogous method is best used as a tool to determine broad price ranges for simple, straightforward projects or as a check to verify estimates prepared using another method.

Communication: Care should be taken to properly communicate with project stakeholders regarding the range of possible cost and schedule changes as the project becomes more defined.

Assumptions: Given the large-scale assumptions inherent in parametric estimating methods, the estimator must document all assumptions clearly.

Uncertainty: Provide an adequate range of cost that reflects the unknowns in the project (see Exhibit 4-2).

Maintenance: Keep the estimate current as the project waits to move into scoping phase.

Tips

1. **Project scope:** Although the project has not been fully defined, do your best to determine the full scope of the project before putting an estimate together. Develop a comprehensive list of all the components that will be required in the project (within the time and resources allowed).
2. **Field review:** "Walking the job" is an excellent way to begin the process of obtaining good quality, factual data to back up the cost estimates and establish confidence in the process. To be confident that the project is adequately planned, perform a systematic field review. Coordinate with the area Maintenance office to have them accompany you on this field review. Many times, maintenance is aware of conditions and issues that may not be readily apparent.

3. A project field review provides an important perspective that supplements the mapping, photos, survey data, and other sources of information about the project that are used in the office. Seeing the proposed project site first-hand minimizes the possibility of overlooking significant design features.
4. While in the field, project personnel should be looking for potential high-cost items (costs of mitigating hazardous waste and other sensitive environmental areas. Document where high-cost items may be present on the project so they can be considered in more detail during project scoping. A strip map and proposed typical sections are very useful in the field to document the occurrence of project features of special interest.
5. **Worst probable case:** Determine and assess in the field the "worse probable case" scenario, particularly on reconstruction projects. Existing facilities previously thought to be adequate may have become inadequate because of changes to standards, new data, etc. Design feature decisions, project constructability, construction staging, lifecycles etc., can often be evaluated and documented in the field in preparation for further study later.

Cautions

1. Market conditions: Unit prices can vary greatly over time based on market conditions. Petroleum products, concrete, and steel prices have probably been the most volatile recently. It is important to review the unit prices, especially for larger-cost items on a project, to reflect current trends. Market conditions are important in all phases, but most prominently in planning since the construction date is so far into the future.
2. Ongoing maintenance: Planning estimates may sit on the shelf for many years before moving to scoping, design, and construction. Planning estimates should be reviewed and refreshed regularly to reflect potential changes in scope, unit prices, regulatory requirements, etc. This is also a good time to review the assumptions and revise as necessary.
3. Precision: The total cost may be displayed by computer estimating tools to the nearest dollar, sometimes even to the penny. When these estimates are shared within the organization, or even outside the organization, they may give the impression that the estimate is very precise. Round planning phase estimates up to the nearest thousand, ten thousand, or hundred thousand dollars, depending on the magnitude of the estimated cost.

4.04 Scoping

Scoping estimates set the baseline cost for the project. A project is programmed when it is entered into the Capital Improvement and Preservation Program (CIPP) and the Biennial Transportation Program. The scoping estimate is important because it is the baseline used by the Legislature to set the budget; future estimates will be compared against it. Clearly document assumptions and scope definitions in the Basis of Estimate document so that changes can be accurately compared to this estimate.

4.04(1) Techniques

Historical bid-based and parametric techniques have been described previously, and both may be used in the scoping phase.

Cost-based estimating uses the same process as historical bid-based estimating, except that the costs applied during estimating are based on analysis of items other than bid research. Historical bid-based and cost-based approaches both utilize quantities estimated for items such as asphalt, concrete pavement, structures, and roadway excavation.

Other items that are not yet quantified may be estimated parametrically or through the use of historical percentages. At this time, risks associated with the calculation of item quantities should be identified, and a Risk Management Plan should be developed and included in the estimating file for future reference.

4.04(2) Considerations

General Advice

Create/Update Basis of Estimate: Clearly document all changes, assumptions, and data sources in the Basis of Estimate. This documentation is crucial because future estimates will reference this work to justify any changes in the cost of the project.

False precision: Do not assume a level of precision that is not warranted at this phase of the estimate. A properly developed estimate will include well-documented assumptions; many of the details that impact project cost are not defined at the time a scoping estimate is prepared. Miscellaneous item allowances at this phase of design typically range from 20% to 30% and may be higher for nonstandard projects (see Exhibit 7-3).

Ensure costs and quantities are rounded to the appropriate significant figure (See Plans Preparation Manual 400.06(5)(f)).

Unit costs: Choose the correct unit costs for items in current dollars.

Major risks: Ensure major risks to the project are identified and documented.

Tips

1. When preparing the baseline estimate in the scoping phase, recognize complexity and the work that needs to be done to minimize uncertainty, such as stakeholder involvement and permits to be documented. If appropriate, call suppliers or specialists to get ideas about cost. Contact city/county engineers to discuss local issues, such as utility work and coordination, and use the expertise of offices within the agency, such as right of way, materials, traffic, environment, and bridge.
2. Once again, a site visit and investigation can be quite valuable. The more familiar an estimator is with the site, the more likely it will be that the estimator will not miss anything while estimating the project. During scoping, all existing structures, utilities, and any other potential obstructions should be noted and quantified. (Look for: costs of mitigating hazardous waste and other environmental impacts; utility relocation; noise barriers; retaining walls; major storm drains; need for a transportation management plan; traffic handling, etc.) Keep in mind that operations and maintenance personnel are aware of problems that are not apparent from a cursory site review.
3. Understand the current design standards and their project impacts. Anticipating potential design changes by staying actively engaged in design development can help while performing a scoping estimate.
4. Define the work as completely as possible. What work will be included in the project? What work is excluded?
5. Identify the major items of work for the type of project being scoped. Focus the efforts on these major items. For example, on a paving project, hitting the cost of striping to the exact penny of the final cost means nothing if the paving quantities are misestimated or the unit cost is way off.
6. Review the project constructability and make sure you are comfortable with constructing the project. Can the project be constructed as planned? Will there be adequate time to complete the project without carrying it into another season beyond what is planned? Are there local contractors who can perform the work? Are there adequate construction staging areas nearby?

Cautions

1. Watch out for traffic problems that will be caused by construction. If detour routes are necessary or available, will the estimate include addressing damage done to the detour route by the added traffic? If detour routes are not available, will daytime lane closures be allowed? If night work is necessary, address it in the estimate.
2. What utility impacts are there? Utility impacts can be direct costs to a project and may result in change orders due to delays when the utility work is concurrent with the highway construction. This holds true for both public and private utilities.
3. Watch out for areas within the project limits with safety performance issues. These areas may need safety improvements that weren't initially identified.
4. The same personnel who originally scoped the project should not be the only ones estimating it. More people and fresh perspectives should always be welcomed.
5. In major improvement projects, inadequate risk planning during scoping can lead to significant challenges. Hidden factors often pose greater risks than visible ones, and the actual scope may far exceed initial expectations. Allocate sufficient time and resources for robust risk management.

4.05 Design

Estimates prepared at various design phases, including Geometric Review, General Plans Review, and Preliminary Contract Review, are used to track changes in the estimated cost to complete the project in relation to the current budget (legislative budget). Each time the estimate is updated, the cost estimating process detailed in Exhibit 2-1 should be followed. Compare the current budget for the project cost and schedule to the new estimate.

Document each update and provide a written explanation of any significant changes to include in the estimate file. If the budget or scope of the project is to be updated, fill out and submit a Project Change Request Form. The final Engineer's Estimate, along with supporting documents, is required to be filed in the Design Documentation Summary (DDS).

At design approval, the scope of work is better defined, and many items are increasingly known. These include right of way needs, permit conditions, environmental mitigation, quantities of major items, and stakeholder needs. As the scope of work becomes stable, confidence in the estimate should improve.

4.05(1) Techniques

Historical bid-based estimating methodologies are typically used for standard bid items known to be included in the project.

Cost-based estimating methodologies can be used for items where there is little or no WSDOT bid history. Cost-based estimating may also be used for major items of work that are known project "cost drivers." Cost-based key resources are suppliers and other individuals knowledgeable about current prices for the subject items, typical construction methodology and production rates, and equipment used. The estimator should contact these resources to develop basic cost data for materials, labor, and equipment.

Risk-based estimating is used to capture the effect of a project's uncertainties on the project cost and schedule. Review risks identified earlier in the project development process and update the Risk Management Plan to reflect the current design phase and risks.

4.05(2) Considerations

General Advice

Guard the estimate against false precision. We often do not know as much about the project as we think we do, particularly early in design. Project definition continues to be refined until the project is ready for advertisement. Use appropriate item allowances for estimates (see Exhibit 7-3).

If cost-based estimating techniques are used, pay special attention to documenting assumptions such as crew size, crew make up, production rates, and equipment mix and type. The costs assumed for contractor overhead and profit as well as for subcontractor work should also be documented. It is important to remember that these assumptions may not reflect the decisions of the individual contractors that will bid the job, thus introducing elements of uncertainty into the estimate.

Tips

1. Early in the project, get rights of entry for all properties adjacent to the project, not only those that seem to be necessary. Repeatedly sending a real estate services employee to obtain these documents is costly and can impact the schedule.
2. Understand the staging needed to construct the project. Work with the proper subject matter experts to verify traffic control strategy. (A subject matter expert or specialty group is a team that has authority in a particular area and profound knowledge or skills in a particular area of endeavor).
3. Do your research. If there are material sources near your project site, you can often get a better price. If you need fill, and another project has excavation to waste, maybe their excavation can be stockpiled for your fill needs (results in haul and placement costs only). Do not forget your local agency projects. Develop a good relationship with the cities and counties you work in.
4. Consider the timing of your advertisement. If you advertise in late fall or winter, your estimate may be lower. Demand is typically lower in these months, resulting in lower prices. Contractors usually have more crews free during this time, and you may end up with their best or most experienced crew for off-season work.
5. Ask the experts. Do not be afraid to call suppliers, specialists, or contractors to get an idea about cost. And do not be afraid to change their numbers to fit your situation (remoteness, limitations, working days, work hours).
6. Consider accessibility to the work zone. How easy is it for the contractor and construction workers to get to and from the work site?

Cautions

- Assuming the scoping estimate and project definitions are correct is not a good way to begin an estimate. When you start an estimate, look with fresh eyes, and identify all substandard conditions. Coordinate with the appropriate entities to determine whether deviations will be approved.
- Not enough contact with functional groups will lead to less accurate estimates. Do not be afraid to ask questions. Spend time getting to know those in functional groups who will be reviewing the project.
- Be sure to pay enough attention to lump sum items. These have the greatest potential for cost overruns.
- Be aware of structures. Work with the functional groups to confirm the estimate for any structures. Reconfirm structure estimate after the geotechnical report is complete.
- Do not take the historical bid data at face value and simply average the prices shown. There are always circumstances that should be considered.

- Avoid dictating the methods the contractor is to use on items that do not require it. It limits the contractor's flexibility and raises your cost. The contractor will find the least expensive way to perform the work.

4.06 PS&E

The Engineer's Estimate is prepared for the Final Contract Review in preparation for advertisement. It is used to obligate construction funds and to evaluate contractors' bids.

4.06(1) Techniques

Historical bid-based, cost-based, and risk-based estimates are the same as Section 4.05. Pay attention to the miscellaneous item allowances (see Exhibit 7-3). At the PS&E phase (letting) the allowances are omitted because the estimator should have all the data needed for the projects. The project data used at this phase shall be identical to the project data provided to potential bidders.

4.06(2) Considerations

General Advice

Carry out the final independent QA/QC checks of calculations, prices, and assumptions during the estimate review. Review the Basis of Estimate for completeness, accuracy, and clarity, and ensure all figures, from detailed backup to summary phases, are traceable.

- Double-check quantities of major items and cost drivers.
- Review specialty group estimates for scope and cost.
- Review Contract Special Provisions.

Tips

1. Do not use any documents for the PS&E (advertisement) estimate that the contractor will not have. If you need additional documentation to complete an estimate, it may be an indication that the plans and specifications are not complete.
2. If possible, use historical data from the region the project will be performed in. These historical bids are probably from the same contractors who will bid on this project.
3. Construction staff can help evaluate the potential impact of staging, materials storage, hauling of materials, location of batch plants, and other constructability-related issues. The information in this step is exceedingly important since the estimate at this phase is completed as though the bidders prepared it.

Cautions

1. Recent drastic changes in material costs or availability may cause a knee-jerk reaction with the contractors while preparing their bids. The WSDOT estimator should think like a contractor while estimating.
2. Previous estimates using elements and categories should now all be reduced to items. Be sure you know everything that was included in the aggregations before splitting them into items. Not doing so is an easy way to miss some items.

Exhibit 4-2 Cost Estimating Matrix

Project Development Phase	Percentage of Design Completed	Purpose of Estimate	Methodology	Tools	Estimate Range
Planning Washington Transportation Plan Highway System Plan Design Studies Route Dev. Plans	0% to 2%	Screening or Feasibility WTP/HSP (20-Year Plan) WTP – Washington Transportation Plan HSP – Highway Systems Plan	Parametric	PLCE	-50% to +200%
Pre-Design	1% to 15%	Concept Study or Feasibility Implementation Plan (10 Yr. Plan)	Parametric Risk-Based	PLCE and Risk assessment models	-40% to 100%
Scoping, Basis of Design (BOD), Project Summary	10% to 30%	Budget Authorization or Control Capital Improvement & Preservation Plan (CIPP)	Parametric Historical Bid-Based Risk-Based	PLCE and UBA, BidTabs.Net Risk assessment models	-30% to +50%
Design Design Documentation I/S Plans for Approval Design Approval	30% to 90%	Design Estimates (Project Control of Scope Schedule Budget)	Historical Bid - Based Cost-Based Risk-Based	UBA, BidTabs.Net, Risk assessment models	-10% to +25%
PS&E Plans, Specs, Estimate (R/W Plans approved)	90% to 100%	Engineer's Estimate (prior to bid)	Historical Bid - Based Cost-Based Risk-Based	Unifier, UBA, BidTabs.Net, Risk assessment models	-5% to +10%

Chapter 5 Cost Estimate Training and Resources

5.01 Training
5.02 Resources

5.01 Training

The links provided below are training sites both within WSDOT and others that provide pertinent training. These links will be updated with available training. Be sure to get supervisor approval before attending training classes.

Cost Risk Assessment– Find guidance and resources on estimating and project risk management.

Development Division Technical Training – Includes Access Control, Contract Plans Review, Practical Design and other topics

Workforce Development – Professional Development, Technical Training, Telework Resources and Wellbeing and Workforce Culture and Diversity, Equity and Inclusion.

Project Management Training – Training on Change Management, Negotiation Essentials, Risk Management, Cost Management and scheduling using Primavera Scheduler.

Training for Fish Passage – Learn how to get involved with fish passage projects, applicable standards and guidance, and view training opportunities

Washington State Learning Center – Training and development tools are also offered through Department of Enterprise Services.

Practical Solutions Training – Training program summarizes WSDOT's planning and project development process and explains the Practical Solutions approach.

Skill Soft – Training includes interactive videos and practical simulations

American Association of State Highway Officials (AASHTO) – Offers a variety of training opportunities as a service to members and the broader transportation industry. From online training and continuing education opportunities to middle-management and senior executive training, AASHTO offers challenging programs that help prepare today's workforce – whether in the hearing room or at the job site. The online AASHTO training is free for WSDOT employees. To sign up, create an AASHTO account (use your WSDOT email address for the username). After creating your AASHTO account, navigate to the technical training and add the desired training to your cart. After that, follow the instructions that are emailed to you.

AASHTO – Practical Guide to Cost Estimating

Association for the Advancement of Cost Engineering International (AACEI) – Access effective online professional development and training on a wide variety of topics. Direct Link to Online Courses.

RS MEANS – Comprehensive construction training courses that fit every schedule and learning style.

University of Washington, Construction Management Certificate – Includes cost estimating course work.

International Cost Estimating and Analysis Association (ICEAA) – Certifications provide professional credentials that set the standard for the entire cost estimating and analysis community.

- Link to ICEAA Certificate Program
- Link to ICEAA Workshops

American Society of Professional Estimators (ASPE) – Serves residential, commercial, and civil construction estimators by providing education, fellowship, and the opportunity for professional development. Direct Link to online courses.

Coursera – Construction Cost Estimating and Cost Control Course.

Management Concepts Project - Cost Estimating Course – This course is intended for intermediate level program and project managers.

WSDOT employees can check with HR to learn more about supported training opportunities.

5.02 Resources

Unifier Design Estimating – Application to provide WSDOT users with a quick, reliable, flexible, and effective way to manage estimates and contract bid information.

Bridge Design Manual (Chapter 12, Appendix 12-A) – Provides WSDOT bridge design engineers with a guide to the design criteria, analysis methods, and detailing procedures for the preparation of highway bridge and structure construction plans, specifications, and estimates.

Policy Statement P 2047 Estimating Project Budget and Uncertainty –

Capital Program Management Systems (CPMS) Tables:

- Construction Cost Index
- PE Inflation Tables
- RW Inflation Table

The above tables are only available to WSDOT personnel.

Federal Highway Administration (FHWA) Major Project Delivery Guidance - Includes Cost Estimating, Financial Plans and Project Management Plans.

Association for the Advancement of Cost Engineering International, AACE International - Provides technical guidance, educational products, events and eight professional certifications to address the diverse need of members and customers.

National Cooperative Highway Research Program (NCHRP) Report 574 - Guidance for Cost Estimation and Management for Highway Projects during Planning, Programming, and Preconstruction

Oman Systems BidTabs.Net (see your IT to set up) - Bid Tabs Pro is software provided by a private company using Bid Tabulation data from our estimates both from EBASE and the Unifier Design Estimating System. It will analyze all Standard Bid Item and contractor bid data for contracts let by WSDOT. It makes this data available with a multitude of predefined query areas and creates reports.

Project Management Guide – WSDOT is committed to project management best practices, this guide offers resources for good practice.

RS Means – Detailed cost tracking for equipment, material, and labor.

Project Risk Analysis Model, PRAM workbook – The Project Risk Analysis Model (PRAM) uses Monte Carlo simulation to generate cost and schedule probability distributions from user input cost, schedule, risk and uncertainty information.

WSDOT Glossary for Cost Risk Estimating Management

WSDOT Contract Ad and Award – Advertisement and Award information for WSDOT projects.

WSDOT Unit Bid Analysis – The Unit Bid Analysis (UBA) contains the bid history for Standard Bid Items used in WSDOT projects.

WSDOT Training Contacts – Find contact information for WSDOT trainers based on the topic of training.

AASHTO Publication Catalog

Chapter 6 Documentation / Basis of Estimate

6.01 Introduction
6.02 Estimate Quality
6.03 Purpose of the Basis of Estimate
6.04 Basis of Estimate

6.01 Introduction

An estimate without documentation is not an estimate. The estimate file should be a well-organized, easy-to-follow history from the first estimate through preparation of the final estimate. The documentation should track changes as the estimate evolves, including updating the scope; assumptions; quantity and price calculations; and risks from the previous estimate.

A well-documented estimate enhances credibility and contributes to transparency and confidence in project estimates. Clear documentation is particularly important as the project passes from one group to another, or as team membership changes. The project estimate file should follow the project through the various stages so that each new estimate can be easily tied to the previous one.

The basis of estimate (BOE) is submitted, along with the estimate and Quality Management Plan, to the Plans Review Office for each plan review.

6.02 Estimate Quality

Estimating time should be scheduled into the project management plan for each phase of the project. This will ensure adequate time and resources are allotted for performing cost estimating.

As part of the estimate review process, external estimate reviews are performed. To ensure the quality of the estimate, someone external of the project team should perform a review of the estimate file. This external review will help ensure the estimator has clearly recorded the assumptions and decisions made in the estimating process.

6.03 Purpose of the Basis of Estimate

The basis of estimate concisely documents all critical aspects of the project cost estimate. This document is used by the project team and management to convey that the estimator has a clear understanding of the purpose of the project and how it addresses the defined needs as the proposed solution to the problem. It also conveys a clear message that all issues have been taken under consideration in the estimated cost of that solution.

6.04 Basis of Estimate

A well-written Basis of Estimate (BOE) clearly and concisely states the purpose of the prepared estimate (cost study, project options, benefit/cost study, funding, etc.); project scope; pricing basis; allowances; assumptions; exclusions; cost and schedule; threats and opportunities; and any deviations from standard practices. The BOE is a documented record of pertinent communications with, and agreements made between the estimator and other project stakeholders.

A well-prepared Basis of Estimate will:

- Document the overall project scope.
- Document the items that are excluded in the project scope.
- Document the key project assumptions.
- Communicate the estimator's knowledge of the project by demonstrating an understanding of scope and schedule as it relates to cost.
- Alert the project team to potential cost and schedule risks (threats and opportunities).
- Provide a record of key communications made during estimate preparation.
- Provide a record of all documents used to prepare the estimate.
- Act as a source of support during dispute resolutions and for bid analysis.
- Establish the initial baseline for scope, quantities, and cost for use in cost trend evaluations throughout the project.
- Provide the historical relationships between estimates throughout the project lifecycle.
- Facilitate the review and validation of the cost estimate.
- Concisely and completely describe the facts.
- Support facts and findings.
- Identify estimating team members and their roles (including Specialty Groups).
- Describe the tools, techniques, estimating methodology, and data used to develop the cost estimate.
- Identify other projects that were referenced or benchmarked during estimate preparation.
- Establish the context of the estimate and support review and validation.
- Be supported by backup documentation
- Qualify any rates or factors that are referenced either in the estimate or BOE; e.g., productivity can be expressed as either units/time (linear feet/hours) or time/units (hours/linear feet).
- Checklist for Estimate Quality.

Chapter 7 Estimate Data, Composition and Markups

7.01 Introduction
7.02 Programmed amount
7.03 Base Cost Estimate
7.04 Base Cost Estimate Markups
7.05 Base Cost Estimate Order of Calculation
Exhibit 7-1 Costs included in the Programmed Amount
Exhibit 7-2 Identified and Quantified Items vs. Miscellaneous Item Allowance in Design
Exhibit 7-3 Markups Summary
Exhibit 7-4 Base Cost Estimate Components (Current Year Dollars, CY$)
Exhibit 7-5 Order of Calculation for Base Cost Estimate

7.01 Introduction

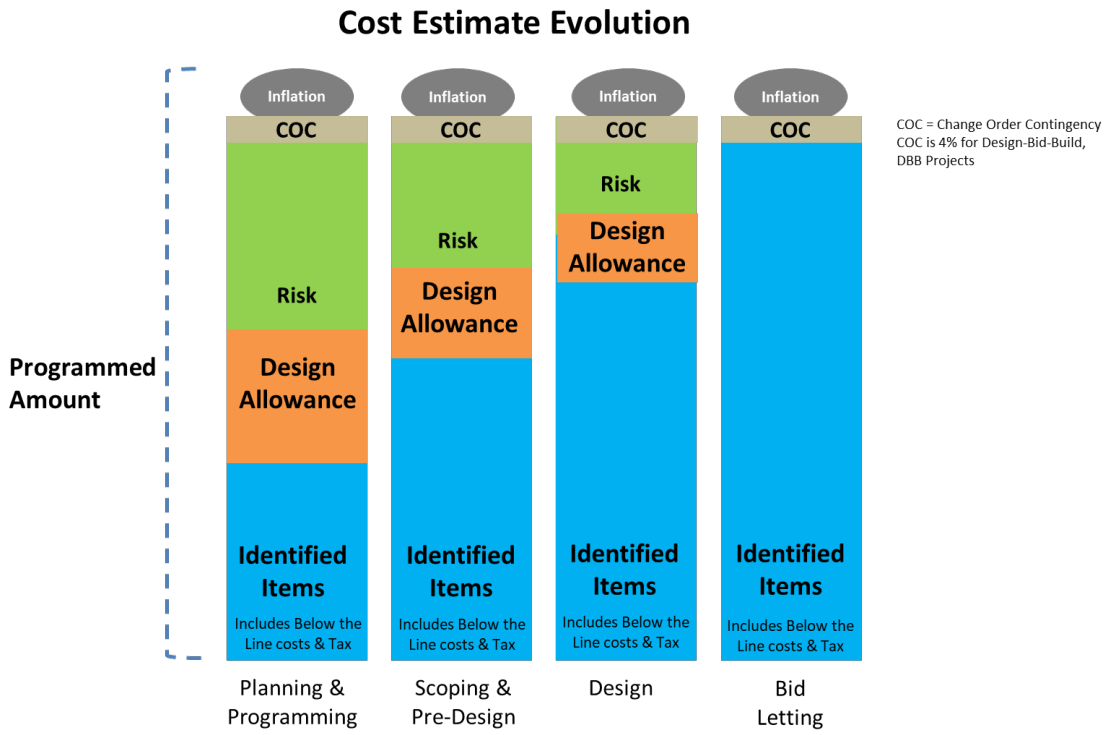
What does a project cost?

There is no single “right” answer for this question. Often it depends on the person asking the question and why they want the information. The amount programmed will be significantly different from the base cost estimate or the project cost estimate.

7.02 Programmed amount

The programmed amount is based on the estimated total dollars required to complete all work associated with the project. This includes all phases of the project; preliminary engineering, right of way and construction as well as project risks, escalation and change orders. See Exhibit 7-1 for a detailed breakdown of the programmed amount estimate.

Exhibit 7-1 Costs included in the Programmed Amount



Unknown Risks - Events that are unforeseen, unidentified, and undetermined. These risks reveal themselves as the project unfolds. Contingency may cover some or all of this.

Escalation - Persistent rise in cost of commodities, goods, or services due to a combination of inflation, supply/demand, and other effects such as environmental and engineering changes, technological changes, environmental effects, political and other effects.

Known Risks - Identified events that are described and include a risk trigger; in addition, these risks are characterized via probability and impact – often this information is produced through formal risk management processes such as project cost risk workshops.

Base Cost Estimate - Total estimated cost of all the project phases: preliminary engineering (design), right-of-way (land that needs to be purchased for the project), construction. An appropriate base cost estimate includes an accompanying basis of estimate, and it has been reviewed for quality and completeness. A project base cost estimate is updated as the project design progresses. It includes known project features, elements, and components. The base cost estimate does not include amounts for risks or uncertainty.

7.03 Base Cost Estimate

The base cost represents the opinion of cost that can reasonably be expected if the project materializes as planned, this includes the total estimated cost of all the project phases: preliminary engineering (design), right-of-way (land that needs to be purchased for the project) and construction. An appropriate base cost estimate includes an accompanying basis of estimate, and it has been reviewed for quality and completeness.

7.04 Base Cost Estimate Markups

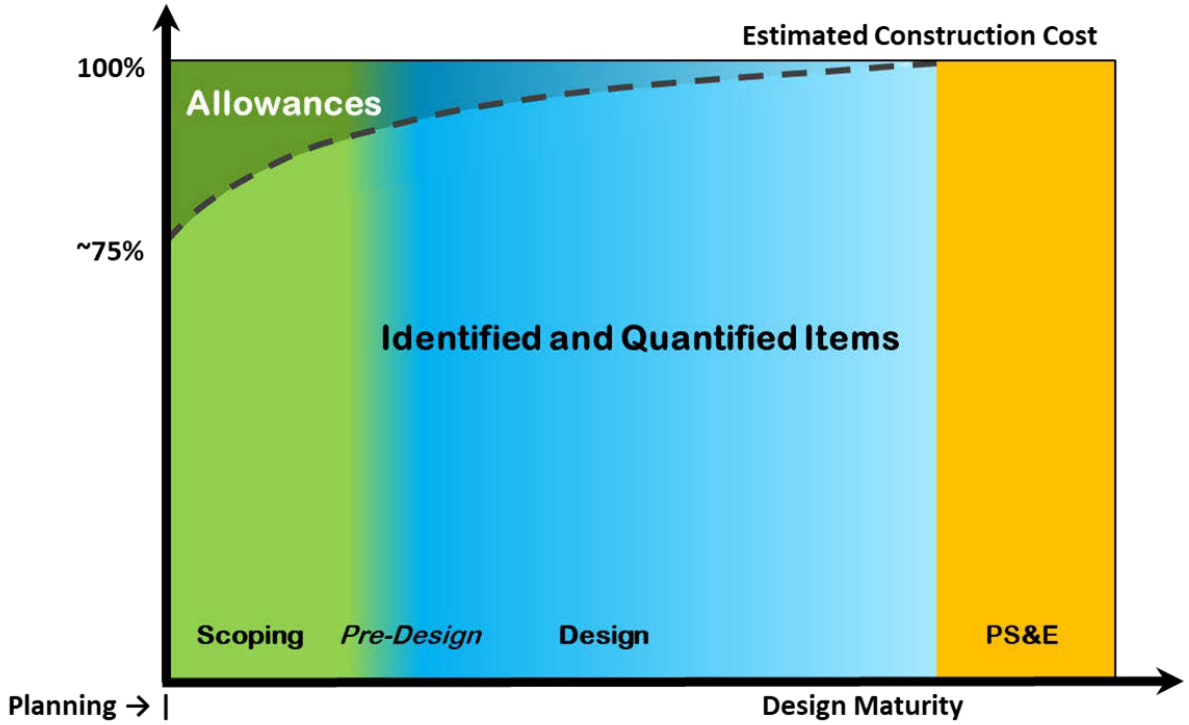
Estimate Mark-ups are percentages applied to the estimated project cost. Mark-ups are generally based on

- Project size
- Location
- Type and complexity of work performed
- Percentage of design completed

WSDOT typically applies six forms of markups in creating the project cost estimate.

Design Allowance – This markup is included as a line item in the design estimate. An early planning level estimate when very little is known about final work items or quantities, may include this item at as much as 50% of the estimated item cost. However, for design-bid-build projects at 100% design (PS&E complete and ad-ready), this markup is reduced to zero. See Exhibit 7-2.

Exhibit 7-2 Identified and Quantified Items vs. Miscellaneous Item Allowance in Design



Design Allowance accounts for lack of scope definition and those items too small to be identified at the stage of project design. This allowance is eliminated for final PS&E estimates on design-bid-build projects because the scope and estimate for all items should be identified at that point. Accounts for known but unquantified items.

Miscellaneous Item Allowance in Design will vary along with advances on design. Exhibit 7-2 illustrates the ideal situation: as items are Identified and Quantified during the Scoping and Design phases, Miscellaneous Item Allowance should correspondingly decrease.

Mobilization – Are the costs for preparatory work and operation including tasks necessary for the movement of personnel, equipment, supplies, and incidentals to the project site, and for all other work and operations which must be performed where costs are incurred prior to beginning work at the project site.

Sales Tax – This markup is based on project location, facility ownership and facility access designation (e.g., managed access, limited access etc.). On WSDOT projects, Control Sections are assigned to various highway segments that govern the application of sales tax.

Change Order Contingency (COC) – This markup represents the construction change order contingency. It is an allowance for unexpected conditions or circumstances encountered post award of the project during construction. Typically, 5% is allowed for all projects regardless of type or size.

Construction Engineering (CE) – This markup is the allowance for construction administration of WSDOT projects. It represents a sum to allow for quality assurance activities of construction personnel such as field inspection, submittal review, etc.

Preliminary Engineering (PE) – Not included in the cost estimate but is based on the estimated construction cost of the project for allocation. See Exhibit 9-1.

The base cost estimate is unbiased and neutral – it is neither optimistic nor conservative. It does not include any risks but does include the WSDOT standard construction change order contingency of 5%.

Exhibit 7-3 Markups Summary

Cost Estimating Elements [1]	Planning	Scoping	Design	PS&E
Miscellaneous Item Allowance in Design	30% to 50%	20% to 30%	10% to 20%	0% (all items should be defined)
Preliminary Engineering	See Exhibit 9-1		PM's Workplan + Actuals to Date	Actual
Mobilization	User Guidance Table may be found in Appendix A			
Sales Tax	Site-specific, based on Control Section. Data can be found in TRIPS or Unifier. Specific direction is found in Standard Specification 1-07.2. Standard Specifications Division 1 or Application of Retail Sales Tax			
Contingency	Applies to parametric, historical bid-based, and cost-based estimates only. See Appendix A Construction Engineering Percentages & Contingencies			
Construction Engineering	Use Guidance Tables found in Appendix A Construction Engineering Percentages & Contingencies			

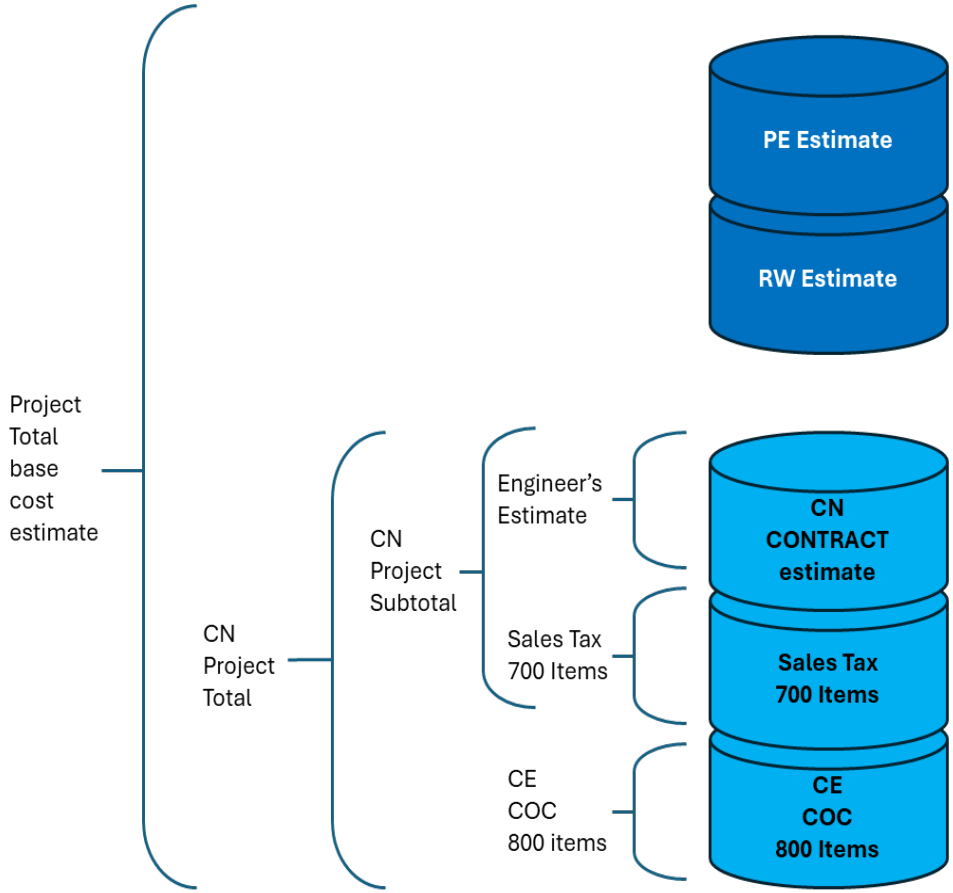
Notes:

[1] Round to appropriate significant digit; for example, $196,526,918.00 is rounded to $197 million.

[2] Report cost estimates in current dollars to Program Management. The Construction Cost Index (CCI) will be used to inflate the estimate to midpoint of construction by Program Management. Note: Public declarations of estimated project costs should be presented in the form of a range in Year-Of-Expenditure (YOE) dollars.

Base cost estimate components are illustrated in Exhibit 7-4.

Exhibit 7-4 Base Cost Estimate Components (Current Year Dollars, CY$)



PE Estimate – The effort (budget/cost) of taking a project from planning through the scoping and design phases of project development. Planning and scoping typically have separate budgets but are encompassed under Design or Preliminary Engineering (PE). The terms "Design" or "Design Phase" are sometimes used Interchangeably with PE.

RW Estimate – The estimated cost to purchase land or easement rights needed for the project.

CN Contract Estimate –Includes: Construction Line-Item Costs, Design Allowance (known but unquantified line items) and Mobilization.

Sales Tax & 700 Level Non-Bid Items – Includes: Engineering Assistant Task Orders, State Patrol and Agreements

Construction Engineering, Change Order Contingency & 800 Level noon-Bid Items – Includes: Construction Engineering dollars, Change Order Contingency dollars, RR Flagging, Emergency Projects and 800 level Non-bid items.

700 and 800 items are called "below the line" items, because they are items of work that the contractor does not bid on or pay for but are part of the overall cost to deliver the contract.

7.05 Base Cost Estimate Order of Calculation

The order of calculation is important in the base cost estimate to ensure that percentage markups such as sales tax, mobilization, construction engineering, preliminary engineering and change order contingencies are applied appropriately.

See Exhibit 7-5 for order of calculation example.

Exhibit 7-5 Order of Calculation for Base Cost Estimate

Example Project		
Bid Item Total =	$1,000,000	
Sales Tax =	8.50%	
Construction Eng. (CE) =	20%	
C.O. Contingencies (COC) =	4%	
Preliminary Engineering (PE) =	15%	
700 level non-bid items	$2,000	← No Tax but, CE and COC applied to these items
800 level non-bid items	$5,000	← Tax, CE and COC NOT applied to these items
860 level non-bid items	$10,000	
Right of Way Acquisition =	$10,500	
To Calculate Base Cost Estimate:		
Add together:		
Bid Item Total	$1,000,000	
Sales Tax ($1,000,000 + 8.50%)	$85,000	
700 level non-bid items	$2,000	
Project Sub Total =	$1,087,000	
Multiply Project Subtotal by CE, COC and PE percentages		
CONSTRUCTION ---------- C.E. 20% X	$1,087,000	= $217,400
COC 4% X	$1,087,000	= 43,000
PLANNING/SCOPING ---------- PE 15% X	$1,087,000	= 163,050
Add Together:		
Project Subtotal	$1,087,000	
Construction Eng. (CE)	$217,000	
C.O. Contingencies (COC)	$43,000	
Preliminary Engineering (PE)	$163,050	
800 level items	$5,000	
860 level items	$10,000	
R/W Acquisition costs	$10,500	
TOTAL BASE COST ESTIMATE	**= $1,536,430**	

Chapter 8 Specialty Groups

8.01 Bridges, Nonstandard Retaining Walls, and Structures
8.02 Right of Way
8.03 Environmental
8.04 Traffic Design and Operations
8.05 Utilities
8.06 Railroad

Specialty groups provide cost estimates for portions of the project relevant to their areas of expertise.

8.01 Bridges, Nonstandard Retaining Walls, and Structures

The costs to construct bridges and non-standard retaining walls and other structures are estimated by the Headquarters (HQ) Bridge and Structures Office.

For planning and scoping phase estimates, contact the Bridge Projects Unit in the HQ Bridge and Structures Office. They will help prepare the estimate using historical bid-based or appropriate available methods and consider structure type, size, and location. For Design and PS&E phase estimates, the Bridge Design Unit will provide a detailed cost estimate for project structures.

Estimators must understand and document what is included and what is not included in estimates from specialty groups. Items not included by the specialty group but required for the project must be included elsewhere in the project estimate. A place to start is the WSDOT Bridge Design Manual. In the WSDOT Bridge Design Manual Chapter 12 and Appendix 12A information on typical costs for many items, production rates and estimated working days are provided. Of course, collaborate with the Bridge and Structures office on your specific project.

8.02 Right of Way

Right of way (ROW) costs vary widely throughout Washington State. Region and/or HQ Real Estate Services should be involved early to help determine appropriate costs for the project. ROW estimates typically include administration, relocation, cleanup costs, and allowances for condemnation.

Real estate markets are best characterized by those familiar with the geographic area. In consideration of this fact, subject matter experts (SMEs) such as Region Real Estate Services and Region Right of Way staff should be asked to provide this estimate. These SMEs can provide input regarding the cost of right of way and the uncertainty associated with the real estate market in the geographic area of the project. Other issues to consider include zoning changes, speculation, growth management plans, and pending comprehensive plan changes.

It is important to get Real Estate Services involved early in the design process, especially if there may be permanent or temporary right of way needs. They can work with the appropriate landowners to facilitate required rights of entry, easements and acquisitions that might otherwise cause costly project delays.

8.03 Environmental

Region and/or environmental staff must be involved in the design and estimating process early. The costs and time durations can be significant for complying with environmental laws and regulations, such as impact analysis, mitigation, permit, public involvement and interagency agreements. Any resulting bid items will need to be included in the project estimate.

8.04 Traffic Design and Operations

Traffic conditions have a significant impact on the project cost estimate. Estimated costs should be adjusted to reflect the complexities and volume of traffic expected during construction in anticipation of the bidding contractors doing the same.

8.05 Utilities

The Utilities Office should also be involved early in the project design phase. Subsurface, buried and overhead utilities are present on a majority of project sites, and conflicts with these utilities may impact a project's cost and schedule dramatically. Unidentified buried utilities may result in costly change orders and project delays. During the Design phase, it is imperative that WSDOT perform their due diligence in identifying all buried utilities through site visits and researching as-built documents.

When a utility is located on an easement and WSDOT acquires the property through ROW acquisitions, WSDOT must pay all relocation costs in addition to providing the affected utility with a new easement. Also note that city-owned utilities that are impacted within the city's limits can add similar costs to a project. Another risk that may affect project cost is triggered when utilities that are not identified during the design phase (due to insufficient engineering investigation or inaccuracy in utility as-builts) are discovered while the project is under construction. Utility conflicts not identified until the construction phase can result in costly change orders and project delays. The Region Utilities Office will help identify any utilities that pose a conflict with the project and prepare any new agreements.

8.06 Railroad

The HQ Railroad Liaison should be involved early in the process when there is railroad infrastructure or property within the project limits. A significant risk that may affect project cost is triggered when impacts to railroads are not identified early in the design phase (due to insufficient engineering investigation, inaccuracy in as-builts or right of way plans, etc.).

Chapter 9 Preliminary Engineering Costs

9.01 Preliminary Engineering Costs

Exhibit 9-1 Preliminary Engineering Percentage for Improvement Projects

9.01 Preliminary Engineering Costs

Delivery of a quality-constructed project begins with quality design. Estimating Preliminary Engineering (PE) is an area of increasing importance and demands the same attention to detail, as do construction estimates and right-of-way estimates.

Early in project development (planning and scoping), PE costs are often estimated as a percentage of the estimated construction cost. PE percentages vary by project type and dollar size of the project. Exhibit 9-1 provides percentage of cost spent on PE based on historical data. PE costs for WSDOT designed projects are about 15% of project cost; this can be used as a starting point if Exhibit 9-1 does not provide a figure.

As the project moves through design, the PE cost should reflect actual costs to date plus the anticipated costs to complete the design work as determined from the project work plan. There is a reasonable minimum fixed cost for procurement, which even small projects incur. If the project cost is under $200,000, it might be combined with another project in order to realize efficiency in contract procurement (Ad, Bid, and Award cost).

For projects that include consultant design, PE costs can be different. The cost of design by consultant is typically more than in-house WSDOT design costs. PE cost estimates should reflect the type of resources (WSDOT and consultant) involved in the design effort.

The Project Management Institute defines a project as “a temporary endeavor undertaken to create a unique product, service, or result. A project is temporary in that it has a defined beginning and end in time, and therefore defined scope and resources. A project is unique in that it is not a routine operation, but a specific set of operations designed to accomplish a singular goal.”

The information in Exhibit 9-1 provides an average cost for preliminary engineering for WSDOT projects. The table is organized by project type and is intended as an initial reference point of cost. A determination should be made on a project-by-project basis if there is a need to increase or decrease the percentage value for preliminary engineering work.

The numbers are to be applied with engineering judgment. The actual percentage used may be adjusted for specific project context and considerations of project size, complexity, location, amount of specialty work, and other relevant factors.

Example

A Rest Area Project, with an estimated construction cost of $3M. It is a routine project on a State Route.

Referring to Exhibit 9-1 page 2, it is seen to be a Preservation Project P-3, subprogram PD.

The table suggests an estimate of 12% for Preliminary Engineering.

You have additional project information; project specific factors include:

- A similar rest area was built on this same route last year.
- The total PE for the project constructed last year was 10.8 %.
- The rest area project you are estimating now is more straightforward and has fewer environmental concerns and no utilities to deal with.

Hence, in your judgment an estimate of 10% PE is appropriate for the rest area project being evaluated.

Be sure to document the estimate and the reasoning for the choice of 10% for estimating PE costs in the Basis of Estimate.

Exhibit 9-1 Preliminary Engineering Percentage for Improvement Projects

Improvement Program PE as % of CN		I1 MOBILITY				I2 SAFETY			I3 ECONOMIC INITIATIVE			I4 ENVIRONMENTAL RETROFIT
Subprogram		IA	IB	IC	IQ	ID	IE	IF	IG	IH	II	IK
		Urban	Rural	Urban Bike Connection	HOV Lane	Collision Reduction	Collision Prevention	All-Weather Highway	Trunk System Completion	New Safety Rest Area	Bridge Restriction	Storm Water Runoff
$ 0	$ 500,000	27%	23%	15%	33%	39%	30%	23%	30%	ND	ND	105%
$ 500,000	$ 1,000,000	36%	23%	15%	30%	38%	30%	20%	30%	ND	ND	80%
$ 1,000,000	$ 2,000,000	38%	23%	15%	30%	38%	15%	20%	25%	ND	ND	50%
$ 2,000,000	$ 5,000,000	34%	20%	15%	25%	25%	12%	20%	25%	ND	ND	30%
$ 5,000,000	$10,000,000	21%	20%	10%	20%	10%	10%	15%	20%	ND	ND	20%
$10,000,000	+++	20%	16%	10%	15%	15%	15%	15%	15%	ND	ND	15%

Notes about the source data % shown is an average of projects within the subprogram; D-B and Emergency projects not included	Number of projects the % is based on.
• 2009 - 2019 projects with a completion date • Cost range is based on final Engineer's Estimate Data Treatment – extreme outliers were eliminated	more than 10
	4 to 9
	1 to 3

Improvement Program PE as % of CN		I4 ENVIRONMENTAL RETROFIT
Subprogram		IL
		Fish Barrier Removal
$ 0	$ 1,000,000	71%
$ 1,000,000	$ 2,000,000	46%
$ 2,000,000	$ 5,000,000	31%
$ 5,000,000	$10,000,000	16%
$10,000,000	+++	10%

Notes about the source data % shown is an average of projects within the subprogram; D-B and Emergency projectsnot included	Number of projectsthe % is based on.
• 2012 - 2022 projects with a completion date • Cost range is based on final Engineer's Estimate	more than 10
	4 to 9
	1 to 3

Preservation Program PE as % of CN		P1 ROADWAY	P2 STRUCTURES		P3 OTHER FACILITIES				
Subprogram		PA	PB	PC	PD	PE	PF	PG	PH
		Paving Safety Restoration	Structure Preservation	Catastrophic Reduction	Rest Area	Unstable Slope	Weigh Station	Program Support	Major Drainage Electrical
$0	$ 500,000	25%	30%	40%	20%	25%	30%	20%	40%
$ 500,000	$ 1,000,000	17%	21%	28%	20%	25%	17%	17%	31%
$ 1,000,000	$ 2,000,000	14%	14%	24%	15%	15%	15%	14%	26%
$ 2,000,000	$ 5,000,000	9%	14%	14%	12%	12%	12%	9%	23%
$ 5,000,000	$ 10,000,000	5%	14%	15%	12%	11%	12%	5%	18%
$ 10,000,000	+++	3%	8%	14%	12%	3%	12%	3%	18%

Notes about the source data percent shown is an average of projects within the subprogram; D-B and Emergency projects not included	Number of projects the % is based on.
• 2009 - 2019 projects with a completion date • Cost range is based on final Engineer's Estimate • Data Treatment – extreme outliers were eliminated	more than 10 4 to 9 1 to 3
No specific data available, percentages were estimated from projects having similar type of construction activities or elements.	similar

Chapter 10 Factors Influencing Estimating Practices

10.01 Geographic Considerations
10.02 Restrictive Work Hours or Method of Work
10.03 Quantity Considerations
10.04 Item Availability
10.05 Scheduling/Lead Time
10.06 Difficult Construction/Site Constraints
10.07 Estimating Lump Sum Items
10.08 Force Account
10.09 Timing of Advertisement
10.10 Inflation
10.11 Expected Competition/Contractor Availability
10.12 Other Contracts
10.13 Specialty Work
10.14 Material Shortages
10.15 Standard Items vs. Nonstandard Items
10.16 First-Time-Use Items
10.17 Soil Conditions
10.18 Permit Conditions
10.19 Allowances
10.20 Construction Change Order Contingencies
10.21 Other Funding Sources / Agreement Work for Others

Exhibit 10-1 Past Inflation (PI) and Future Inflation (FI)

Major factors are various items associated with a project that can have an impact, either minimally or significantly, on the project cost estimate development.

The estimator must accurately understand and document the impacts that these Major factors are anticipated to have on the cost estimate.

The following are common Major factors that must be examined to determine whether they influence a project's cost estimate.

10.01 Geographic Considerations

Geographic considerations are powerful project characteristics that may substantially affect the selection of unit bid prices. The project's location, whether in an urban, suburban, or rural setting, should be considered in establishing bid prices. Some of the cost considerations relating to a project's location may be accounted for in the mobilization bid item, dependent on the restrictions of Standard Specification 2-01.

A project in an urban setting generally has to contend with construction operations occurring in more confined workspaces and limited equipment staging, with greater volumes of traffic, limited hours of operation, nighttime work, etc. Some of these factors may be offset by the availability of local contractors, materials, equipment, and labor.

Projects located in rural settings have factors that affect the establishment of unit bid prices different from the projects located in urban settings. Construction operations may have less restricted work areas, less traffic to contend with, and additional hours to complete the work—all factors that increase productivity. On the other hand, materials, equipment, and labor may have to be brought in from out of the area, which may increase costs related to transportation, support, wages, per diem, etc.

For projects that utilize large quantities of aggregate, whether for base, surfacing, or earthwork, the location of material sources and disposal sites may have a large impact on costs. Nearby material sources or disposal sites reduce hauling costs. On rural projects, the cost of bringing in a concrete batch plant, hot mix asphalt plant, or similar facility, may increase unit bid prices. Again, dependent on the restrictions of Standard Specification 2-01, those costs may be directly attributed to and reflected in the mobilization bid item.

Terrain may also be a consideration in establishing an item's cost. Mountainous terrain and steep grades cause production rates to fall, whereas level terrain and straight roadways generally have the opposite effect.

Groundwater conditions can vary greatly and need to be investigated to determine the extent of dewatering required for foundations and other structures such as stormwater retention ponds. Varying geotechnical conditions are covered in Section 10.17, Soil Conditions.

Material hauling that must be accomplished by entering and exiting only on interchange ramps, hauling uphill rather than downhill, working on the top of slopes or retaining walls, etc., is always more expensive to construct than work that is easy to accomplish on level or gentle slopes. The ease of accessibility to the work will affect the cost to do the work.

10.02 Restrictive Work Hours or Method of Work

Restricting the contractor's working hours or the method of work on a project may have major effects on prices. The prices for work that is limited to short shifts, or required to be completed in long shifts, or limited to nighttime operations should be increased to reflect the cost of premium wages required for such work and for the general inefficiencies and decreased productivity that may result. Night work for plant operations (e.g., asphalt concrete production) can be especially expensive when small quantities are involved. Plants usually do not operate at night and may require special production runs at much higher-than-normal operating costs.

10.03 Quantity Considerations

The quantity of a given material on a project impacts the unit cost of constructing and/or supplying that item. This is not simply a supply and demand issue, but also one of production efficiency and economy of scale.

- On one hand, the unit price for larger quantities of a given material will be less than for smaller quantities. Larger quantities may give rise to efficiency by gained experience and the expertise of the contractor's personnel in completing the work.
- On the other hand, for some projects, very large quantities of certain materials may cause an increase to the unit bid price. For example, a project with numerous or large structures may affect the market for a particular type of steel or the availability of cement or may even tie up a region's labor resources.

Small quantities of items of work are often less cost-effective to construct and lead to higher unit prices. Supplier's charge more for smaller purchases, and in some instances the minimum amount that has to be purchased is greater than needed for the project. Smaller- quantity items may be subcontracted out; this increases a contractor's overhead and usually results in a markup being applied to those items.

10.03(1) Separate Operations

Separate operations will usually increase item costs, especially if the order of work or the work unit is to be constructed in scattered locations throughout the project (each requiring move-in and move-out costs). If this is the case, unit prices should be based upon the smaller unit sizes and should not be based upon the entire quantity for the total project.

10.03(2) Handwork and Inefficient Operations

Handwork and small or inefficient operations (even though equipment may be used) will have higher unit costs than work that is able to be mass produced or constructed by using techniques that result in higher production rates.

10.04 Item Availability

Materials that are readily available, or ones that are commonly used, are generally less expensive to purchase and install/construct. Materials that are in short supply are more expensive. This should be considered in establishing the unit price.

Large quantities of materials required in a short period of time may result in a temporary shortfall in product availability and potential cost increases or delays to a project.

10.05 Scheduling/Lead Time

To be efficient, a contractor needs to optimize the scheduling of resources, including labor, equipment, and materials. The ability to plan for and maximize resources allows contractors to be more competitive in bidding. Lead times need to be considered in estimating the project based upon when it is expected to be built. For example, a project that is two seasons long may have the majority of its paving in either the first or the second year.

10.06 Difficult Construction/Site Constraints

Difficult construction and site constraints will increase the cost of construction for a contractor. Examples of site constraints that should be considered include: placing piles under water; working near railroads or nearby historical buildings (possibly fragile); constructing on or near culturally important or environmentally hazardous sites; and working in areas with limited room to construct an item.

10.07 Estimating Lump Sum Items

Lump sum bid items can be more difficult to estimate. They can reduce administrative costs in contract administration, as well as allow a contractor a bit more freedom to be innovative with regard to work means and methods. Lump sum items transfer the risk of performance and quantities to the contractor. If the work to be performed can be quantified, then a payment method that includes a quantity should be used.

Lump sum bid items may also be used when an item of work can only be defined in general terms, such as when the finished product can be defined but not all the components or details can be easily determined. This can make estimating lump sum items difficult for the estimator. It is recommended that the lump sum item be defined as detailed as feasible in order to accomplish a reasonable lump sum estimate.

An estimator should define a lump sum item in terms of its simplest, most basic components and should also consider other factors that may not be easily estimated. By breaking out a lump sum item into smaller items of work that have historical data, and then applying reasonable estimated prices to those subunits, the estimator can calculate the overall lump sum item. Contractors cannot necessarily rely on overruns to cover work that they did not foresee and may adjust the price upward to take on this risk. Therefore, the use of lump sum items should be used with great care.

In some situations, as may be the case for a time-based lump sum bid item such as project temporary traffic control, the lump sum payment may provide the incentive to perform the work more quickly. In such situations, hourly pay items offer no incentive, and may result in the contractor remaining in the work zone for a longer period of time.

Lump sum items should only be used when the following conditions apply:

1. The lump sum item is a bid item with no appropriate alternative non-lump sum standard item available for use.
2. The work is not easily defined. In other words, the final product is known, but the construction techniques or other components are difficult to determine.
3. The project has complex items with many components (although the designer is encouraged to break down constituent items if possible).
4. The lump sum payment may be justified as incentive to complete the work in a more timely or efficient manner than if other units of measure were used.
5. The lump sum item may be justified as less expensive than a force account item (see below) or where the risk assumed by the contractor is low.

The use of a lump sum item must be justified, and the work breakdown documented in the estimate file..

10.08 Force Account

Force account is a method of payment that pays the contractor the actual expenses for all labor, materials, and equipment to complete the work. Markups for material costs, labor surcharges, overhead, and profit are usually added to this figure. This method of payment is used primarily for “extra work” (i.e., work that is unforeseen at the time a project is let or advertised and is discovered during construction) or for items of work that are poorly defined and may or may not be used during construction.

The case of poorly defined items of work is the one most frequently encountered by the estimator. Since the contractor does not usually bid on this work, there is little incentive to reduce costs or execute the work diligently. Because of this, the force account method of payment should only be used when necessary.

When a reasonable cost estimate for a force account item is required, the estimator should try to establish the scope of work to be accomplished. Once the scope is developed, it can be compared to historical bid price data for similar items of work.

If no comparable history exists, the force account item should be broken out into its anticipated core components. The estimator can then rely on historical bid data for those items and the given limitations to come up with a reasonable force account estimate. If no such data exists, the estimator may need to estimate the amount and costs of materials, equipment, and labor to execute the work.

Force account may also be considered as a tool for transferring risk from the contractor to the owner. If the work is properly directed by the project inspector, force account may actually cost less than an equivalent lump sum item.

Use of force account items and their estimated costs must be documented and justified in the project estimate file.

10.09 Timing of Advertisement

The timing of a project advertisement can influence bid prices. Contractors can have a time of year that is busier than others. This relates to weather and construction windows; for example, asphalt paving is not done during cold, wet weather. The appropriate timing of advertisement can also be affected by other items like fish, migrating bird or HMA paving windows and other outside constraints.

The peak season for most highway projects is April through October. The peak season for marine work is November through March. The best time to advertise a project is several months before the work season for

that type of work, to allow time for contract execution and for contractors to mobilize their resources in time to take advantage of the full work season.

If contractors have fully allocated resources for the upcoming season, they are less likely to bid on a project and, if they do bid, it is in a less competitive environment.

For this reason, there is a benefit to WSDOT to advertise a project as soon as possible prior to the peak season, to allow contractors time to plan, schedule, and seek as many opportunities as possible to find efficiencies in their work plans. Hopefully, this promotes a more competitive bidding environment.

The estimator preparing the final Engineer's Estimate needs to be aware of the timing of the advertisement. The estimator should account for any expected fluctuations in bid prices due to the season, such as lower production during temperature extremes, additional protections for weather-sensitive materials, and so forth.

It is important to know if the work can be accomplished before winter weather causes the project to shut down. If the job cannot be finished before the end of the construction season, and the project needs to be suspended, contractors will increase their bid prices to cover their overhead during the winter and repair any damage that may occur. Even if contractors reasonably expect to finish before the winter, they may protect themselves to allow for an early winter.

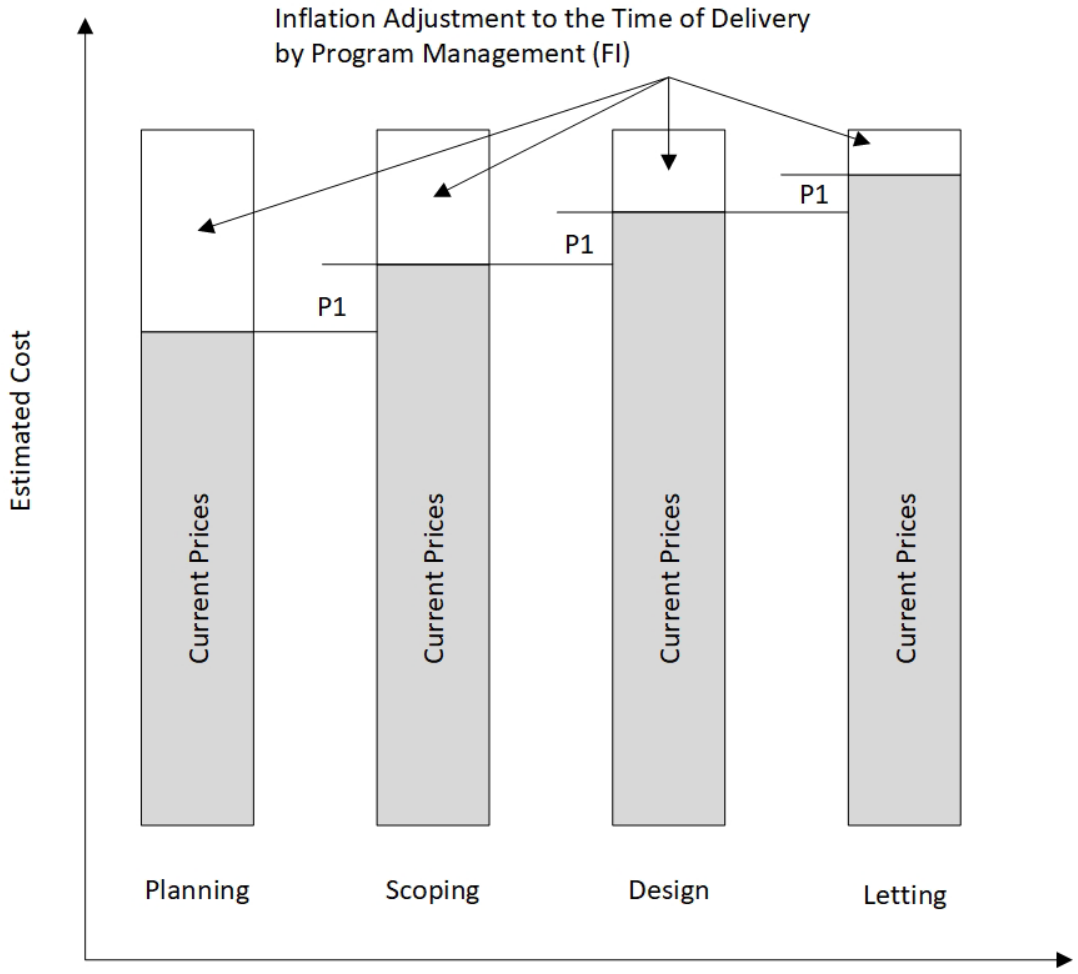
10.10 Inflation

The estimate represents the probable cost at the time of estimate. Any time an estimate is created or updated; the unit prices must reflect the cost at the time the estimate is done. It may require the estimator to go back to the unit price database and get the latest information.

The estimator must always adjust the unit price for inflation considering the interval of time from when the price was used and the estimating time.

Exhibit 10-1 illustrates the ideal situation, when the current prices representing the cost at the time of estimate increase because of the Past Inflation (PI) applied to the prior prices. Simultaneously, the figure shows that the estimated amount created by Future Inflation (FI) (what program management does) decreases since the time span decreases. WSDOT CCI, RW, and PE inflation tables may be obtained from CPDM and are only available to WSDOT employees.

Exhibit 10-1 Past Inflation (PI) and Future Inflation (FI)



10.11 Expected Competition/Contractor Availability

Projects that are advertised for bids late in the season or after contractors have scheduled their work for the year, can expect higher bid prices. This is due to the lack of competition or contractor availability. Projects that are bid during a period of time when a large number of contractors are available may be bid more competitively.

Competition could be a major factor of the cost of the construction. When the estimator expects just one bidder, then the project cost may be much higher than otherwise. On the other hand, if there are many bids, the project cost may be significantly lower than the estimate.

10.12 Other Contracts

Multiple projects being advertised at the same time can influence bid prices in much the same way as lack of competition and availability. The contractors only have so many resources available to develop bids for projects. In the case of large projects, a contractor may not have the resources to develop bids for more than one project at a time.

The most prudent course of action in this case is to manage the program of projects to ensure this does not become an influencing factor on the bids. If this cannot be prevented, then the estimate needs to reflect that multiple bids will be developed at the same time.

The estimator should consider to what extent the reduction below the normal number of bidders will influence the bid amount. As competition decreases, prices increase—hence if the number of anticipated bidders is lower than normal, the bid amounts are expected to increase.

The probability of the occurrence of this risk should be evaluated by the estimator. Common mitigation strategies include timing of the advertisement and work packaging. On the other hand, in extremely competitive bidding environments, with large numbers of contractors bidding, the bids may actually come in lower than expected.

Another factor to consider is that multiple active projects can create conflicts in an area. For example, multiple large-scale bridge projects in a given area may create a shortage in structural steel or skilled labor. In these cases, the estimator must be aware of the ability of the market to support multiple projects.

Other conflicts could include traffic control, labor issues, and direct coordination issues, among others. These conflicts need to be considered in the calculation of production rates and subsequent bid item prices. Project managers should be aware of adjoining projects and nearby work (even from other regions or local agencies). There may be opportunities for collaboration and coordination that will result in more competitive bids and better maintenance of traffic.

10.13 Specialty Work

Specialty work is not necessarily unique items or new construction methods. Rather, it is work that is somehow different than the majority of the work on a given project. For example, on a pavement rehabilitation project, the signal work may be classified as specialty work, although it would not be classified as such on a project that was predominantly signal and lighting work. Estimating the cost of specialty work requires a thorough understanding of the work involved and the resources required for accomplishing it.

When estimating specialty work you are not familiar with, seek the advice of experts who can help. Consider the number of qualified contractors capable of doing the project or elements of work.

10.14 Material Shortages

Material shortages will have a major effect on prices since prices are directly affected by supply and demand. Where a shortage is especially acute, a change in design might be considered rather than increasing prices.

10.15 Standard Items vs. Nonstandard Items

Standard items, as listed on the WSDOT Standard Item Table, are familiar to both WSDOT and the contractors. These items of work typically represent a known quantity and quality to both WSDOT and the contractor, and bid history tends to reflect that. When an item is changed in some way to become a nonstandard item, then uncertainty is introduced. This uncertainty typically results in an increased price for the item, especially the first-time contractors see it in a contract. Typical practice should be to use standard items whenever possible. When a standard item is changed and becomes a nonstandard item, the estimator should recognize that the price may differ from the historical prices.

10.16 First-Time-Use Items

On occasion, items of work are included in a project for which WSDOT has little or no historical data to establish unit prices. In these instances, similar items may provide some guidance, but additional investigative work may be necessary.

If the item is thought to be of minor significance, there may be little benefit in spending much time in determining a reasonable bid price. If the item is considered major or is likely to be significant to the overall project bid, research should be conducted to establish a cost. Contacting others who are familiar with use of the item can usually help in determining the cost.

Suppliers, other state departments of transportation, the HQ Engineering Analysis Office (EAO), Regional Transportation Commissions, Port Authorities, Consultant Letting, RS Means Publications, and contractors can be valuable resources in establishing costs. Be wary of relying on estimates from a single contractor or source. Multiple sources should be utilized in developing an estimate for first time use items.

If the item in question is unique in some manner (innovative, or experimental), or it is considered a specialty work, costs may need to be adjusted to account for the contractor's lack of experience with it and the potential increased risk in construction. If the work is likely to be subcontracted out, then the prime contractor may also add a markup to the subcontractor's price.

10.17 Soil Conditions

General assumptions about soil conditions may be made early in the estimating process, but they may turn out to be wrong. As the estimate progresses, geotechnical data may help improve the information and prevent costly change orders and claims. In the early estimates, the assumptions regarding soil conditions and the potential effects of unknown soil conditions should be clearly documented.

A common estimate omission is accounting for shrink and swell of material. The Region Materials Engineer should be consulted to determine the appropriate shrink or swell factor to use. Soil conditions can be a significant cost risk to a project. Risk-based estimating techniques should be utilized to quantify geotechnical risks if they pose a significant threat or opportunity.

10.18 Permit Conditions

Throughout the stages of planning, scoping, and design, various projections of permit conditions for construction can be obtained from Region or HQ Environmental Offices. Engaging these groups early helps to identify specific permits or conditions that can drive up construction costs and identify opportunities to avoid costly environmental conflicts.

At the same time, the estimate must address:

- The actual out-of-pocket costs for developing and constructing transportation projects.
- The labor, material usage, equipment usage, and travel expenses of staff or consultants doing environmental analysis and documentation during project development.
- The purchase of ROW, materials, and installation labor, and additional purchases for major environmental mitigation features such as wetlands restoration and noise walls during construction.

Be aware that considerable costs may be required due to stormwater collection and treatment; wetland protection and mitigation; hazardous materials testing; containment and treatment; and removal and disposal of underground fuel tanks, creosote timbers, and contaminated soils.

10.19 Allowances

Allowances cover items known to be needed for the project yet not specifically identified and/or quantified. Allowances are different from construction change order contingency. Allowances account for a lack of project definition during the preparation of estimates.

In WSDOT estimates, allowances should be shown as a separate line item in the base cost estimate. Allowance amounts should be identified with an explanation as to what they are intended to cover. This is so they can be managed and reduced appropriately as the design progresses and will assist in review of the estimate.

Exhibit 7-1 illustrates the negative correlation between the cost of allowances and the cost of identified and quantified items. Once identified and quantified items are added to the base estimate, the allowances should be reduced in order to reflect the new data available.

10.20 Construction Change Order Contingencies

Construction change order contingencies are typically meant to cover a variety of possible risks or events that are not specifically identified or quantified, such as uncertainties in quantities and minor risk events related to quantities, work elements, or other project requirements during construction. Refer to the Plans Preparation Manual, Section 800.3(2), for guidance on estimating construction change order contingencies.

10.21 Other Funding Sources / Agreement Work for Others

Funding sources should be documented in the Basis of Estimate. Federal funding, participation from local agencies (e.g., participation in intersection improvements), or funding from public/private partnerships should all be documented.

Chapter 11 Independent Cost Estimate Review

11.01 Introduction
11.02 Internal Project Team
11.03 Peer Review
11.04 Region/Headquarters Review
11.05 External

Exhibit 11-1 Activity/Responsibility Matrix

11.01 Introduction

Each estimate should have some level of review, as indicated on the flow chart provided in Exhibit 2-1. Project complexity is an important driver of the level of estimate review. The level of review should be carefully chosen by the project manager.

Estimates are key outputs of the project management process and are fundamental documents upon which key management decisions are based. It is recommended, and may become policy at a region's discretion, that all cost estimates be independently reviewed by estimating staff specialists, subject matter experts, and others as appropriate. Estimates will then be reconciled and revised as needed to respond to independent reviewer comments.

Independent Cost Estimates (ICEs) are third-party evaluations of project costs conducted separately from internal agency estimates. ICEs can serve as a mechanism to promote transparency, accuracy, and accountability in the planning, budgeting, and procurement of civil infrastructure projects.

Owner-agencies should consider commissioning ICEs under the following circumstances:

- Prior to major funding decisions or contract awards
- For validating contractor proposals or change orders
- For high-risk, high-cost, or technically complex scopes

ICEs typically provide:

- **Objective cost assessments** based on industry-standard data and methodologies
- **Expert insight** into construction means, methods, and assumptions
- **Schedule evaluations** informed by production rates and sequencing logic
- **Benchmarking** against similar projects and market conditions

Agencies should ensure that ICE providers are qualified, independent, and possess relevant experience in the applicable infrastructure domain. The scope of the ICE should be clearly defined, and its findings documented in the project record.

In the event of a significant difference of opinion, an estimate reconciliation meeting will be held, and the results documented. The final results of the independent review and reconciliation meeting must receive management endorsement before any project is advertised. See chapter 2.08 Management Endorsement

Independent reviews (check estimates) should be made by experienced estimators who are familiar with the type of work inherent in the project, and who have had no involvement in the development of the project estimate to date. The independent estimator consults with other independent sources such as design engineers, construction managers, or other estimators as needed on specialty items of work.

Checks performed by independent estimators include:

- Reviewing the estimate file and Basis of Estimate document for completeness and readability.
- Ensuring the name(s) of estimator(s) involved in preparing the estimate are shown.
- Ensuring the estimating methodologies are noted by individual items of work.
- Reviewing the overall estimate documentation to ensure it is clear and the figures are traceable, from detailed backup to summary phases.
- Conducting a detailed check of the estimate to include:
    - Checking the development of unit rates and quantities of those items that drive the majority of the bottom-line cost (typically 20% of the items that comprise 80% of the estimated project cost).
    - Making notes of comments on unit rates and quantities.
    - Checking for mathematical errors.

11.02 Internal Project Team

The internal project team review should include a quality control check of quantities and prices. It should also include a quality assurance check that the proper procedures were followed, the documents are complete and clearly understandable, and the final costs and schedule are deemed reasonable for the project scope, size, location, and complexity. The advantages of the internal project team review are that the project team's schedule is easier to coordinate than outside resources and the reviewers have a base knowledge of the project.

The internal reviews tend to be conducted with the same vision, framework, and assumptions as the project teams and this may introduce unwanted bias. This can lead to a review that does not objectively assess all the parameters that affect the estimate.

One useful double-check is to prepare estimates in two different ways and then compare them.

For example, if a project team utilizes historical bid-based methods for preparing the project cost estimate, then consider selecting the top five to ten items in terms of cost and estimate them using a cost-based approach (materials, equipment, and labor). The comparison of the two estimates may be enlightening.

11.03 Peer Review

This type of review is similar to the internal project team review, but it uses another project team or office to conduct the review. Offices can often perform reviews for each other in this way. The advantage to this type of review over the internal review is that the project gets a review with a fresh perspective on the estimate. This can provide an increased level of confidence in the estimate. This is also a good way to share lessons learned and information between project offices and serves to efficiently utilize fixed engineering resources within the department.

11.04 Region/Headquarters Review

This type of review at the region and Headquarters level is typically more formal. One advantage to this type of review is that the reviewers are external to the project team and can thus provide a truly independent perspective on the project. However, the biggest advantage is that the reviewers typically have significantly more experience in performing this type of independent review. Region reviews can be coordinated with region staff.

Headquarters reviews should be coordinated with Engineering Analysis Office staff.

11.05 External

In this type of review, external experts are brought in to review either specific pieces of the estimate or the entire estimate. This type of review can be combined with any of the other types of reviews to supplement knowledge of a specific item of work or to provide an outside perspective. This type of review has a wide range of costs but can provide significant confidence in the estimate that might not be otherwise attainable from internal WSDOT resources.

Exhibit 11-1 is a matrix that identifies which position has responsibility for reviewing and endorsing each type of WSDOT estimate.

Exhibit 11-1 Activity/Responsibility Matrix

Changes – Scope Changes, Schedule Changes, Budget Requests	**PE/PM through Region PDE or EM Must Process Control Change Form through the PC&R Office**
Planning Phase	Region Planning Manager
Scoping Phase, including: Design Documentation Package (DDP)	Region Plans Office and Programming
Design Approval Design Report Alternative Analysis	ASDE, PE/PM, PDE/EM, and Region Plans Office
Determine Contingencies	PE/PM, PDE/EM
PS&E Phase Estimates	Estimator, Designer, and PE/PM
Documentation, including: Assumptions Quantities and Adjustments Prices and Adjustments	Estimator, Designer, and PE/PM
Review Estimates	Region Plans Office and Peer Review Team
Check Estimates and Calculations	Region Plans Office
Prepare Engineer's Estimate	Estimator, Designer, and PE/PM
Independent Estimate	To be determined
ASDE – Assistant State Design Engineer PC&R – Project Control and Reporting PDE – Project Development Engineer	EM – Engineering Manager PM – Project Manager PE – Project Engineer

Note:

The Region Project Engineer, Project Manager, Project Development Engineer, and Engineering Manager should be informed of changes and updates to project scope, schedule, and cost estimate.

Chapter 12 Design Build Estimating

12.01 Introduction
12.02 Design – Build Estimate Considerations
12.03 Preliminary Engineering
12.04 Construction Engineering – Owner / Design Builder
12.05 Contract Contingencies
12.06 Right-of-Way
12.07 External Third Party (e.g., Utilities and Railroad Adjustments)
12.08 Transportation Demand Management and Transportation System Management
12.09 Risk and Uncertainty
12.10 Allocation of Risk
12.11 Value Engineering, VE
12.12 The Engineer's Estimate for a Design Build Project
12.13 Bidding Climate Impact
12.14 Price Adjustment in Contract Provisions
12.15 Highly Specialized Designs
12.16 Construction Time
12.17 Construction Incentives
12.18 Number of Concurrent Contracts and Contract
12.19 Transportation Management Plans for Work Zones
12.20 Environmental Impacts

Exhibit 12-1 Costs included in the Programmed Amount, DB

12.01 Introduction

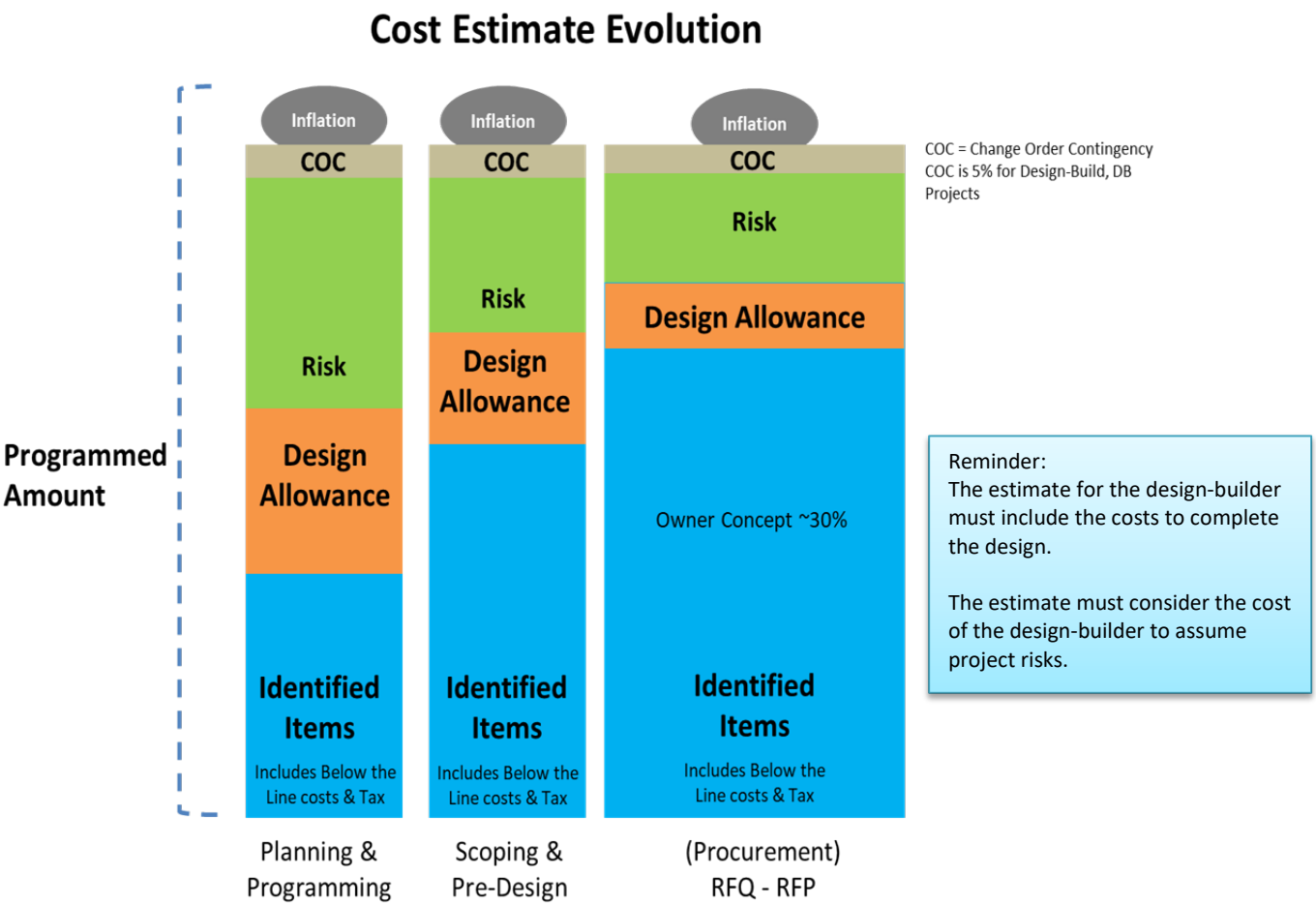
In both Design-Build (DB) and Design-Bid-Build (DBB), early cost estimating focuses on feasibility and funding, relying on conceptual scopes and limited data. As projects advance, DB shifts toward performance-based scoping to support RFP development, while DBB progresses through detailed pre-design to produce complete Plans, Specifications, and Estimates (PS&E).

In DB, the proposer completes project design and construction—offering flexibility that can foster innovation. DBB, by contrast, completes design prior to bidding, following a more linear and prescriptive path.

At the time of advertisement, DB estimates are conceptual and include risk. DBB estimates are detailed and itemized, based on finalized design documents. DB emphasizes adaptability and integration; DBB prioritizes control and precision. As a DB estimate is prepared for procurement it's important to include preliminary engineering costs and account for lump sum proposal items governed by Tax Rules **170** and **171**. Typical bid items for a design-build contract can be found in the resources section, in the 'Contract Form-DBM Guidance' document, on the **HQ Design-Build SharePoint** site: **Design-Build Resources**.

Additional DB costs may include utility relocations (if within scope), technical support, inspection, Design-Builder stipends, third-party coordination, and risk contingencies. The programmed amount should reflect the total cost to deliver the project, including preliminary engineering, right-of-way, construction, escalation, and change orders. The Design-Builder's estimate must also cover the full cost of completing the design. For a detailed breakdown, see Exhibit 12-1.

Exhibit 12-1 Costs included in the Programmed Amount, DB



Unknown Risks - Events that are unforeseen, unidentified, and undetermined. These risks reveal themselves as the project unfolds.

Escalation - Persistent rise in cost of commodities, goods, or services due to a combination of inflation, supply/demand, and other effects such as environmental and engineering changes, technological changes, environmental effects and political effects.

Known Risks - Identified events that are described in the risk register and include a risk trigger; in addition, these risks are characterized via probability and impact – often this information is produced through formal risk management processes such as project cost risk workshops. M22-01.21 Design Manual (wa.gov)

Allowances - Additional resources included in an estimate to cover the cost of known but undefined requirements for an activity or work item. Allowance is a base cost item.

Base Cost Estimate - Total estimated cost of all the project phases: preliminary engineering (design), right-of-way (land that needs to be purchased for the project), and construction. An appropriate base cost estimate includes an accompanying basis of estimate, and it has been reviewed for quality and completeness. A project base cost estimate is updated as the project design progresses. It includes known project features, elements, and components. The base cost estimate does not include amounts for risks or uncertainty, and should be free of optimistic and pessimistic bias.

Programmed amount - Total funding assigned to a transportation project or phase within the Statewide Transportation Improvement Program (STIP). It reflects the formal commitment of resources for that project over a four-year period

Engineer's Estimate - The estimate at time of bid, but also used by some to mean any estimate done during the Preliminary Engineering phase.

12.02 Design – Build Estimate Considerations

As stated previously, creating the design-build estimate is similar to developing the design bid build estimate, but it must include preliminary engineering costs in the overall estimate. Estimates are prepared in Current Year dollars and then inflated to Year of Expenditure (midpoint of activity) by CPDM. Considerations include:

- Basis of estimate and organized documentation.
- Risk-based assessments
- Independent validation.
- Project scope.
- Initial preliminary engineering for owner concept design (design allowance).
- Design Builder costs to complete final design.
- Design Allowances
- Right-of-way and administrative costs.
- Third party (e.g., utility, railroad) costs.
- Transportation Demand Management / System Management (TDM/TSM) costs.
    - (i.e. ramp metering, ATMS)
- All construction costs.
- Construction contingencies.
- Construction administration.
- Public outreach cost.
- Stipend amount (paid under the PE phase)
- Design Builder cost to maintain the roadway etc. during construction.
- Other construction costs – asphalt adjustments, incentives, payable agreements, etc.
- Seek industry input (e.g. AGC/WSDOT Teams, or other)

The following sections are detailed explanations of some cost elements. These should always be included when preparing a program cost estimate for a major project.

12.03 Preliminary Engineering

The cost to the owner to prepare the conceptual design and contract documents is estimated. It may include field investigation, geotechnical borings (depending on the project, this could be significant, with access and potentially right-of-way needs), testing, and administration of the design work and contract procurement, including support offices. It also includes the cost of the NEPA and other environmental and permitting documentation. All costs should be considered including administrative costs through contract execution.

The cost of a General Engineering Consultant (GEC), for this work, if applicable, is included here.

All Preliminary Engineering activities that must be accomplished by the owner in a traditional Design Bid Build project also need to be accomplished or completed in the Design Build contract. However, only the Preliminary Engineering activities performed by WSDOT or WSDOT's consultant should be included as Preliminary Engineering for a Design-Build estimate.

The Design Builder is responsible for advancing the project from conceptual design through final design.

The Design Builder's engineering responsibilities may include, but are not limited to:

- **Completion of the final design** in accordance with project requirements and standards
- **Preparation of detailed construction cost estimates** to support budgeting and procurement
- **Value engineering and risk analysis** to optimize project performance and mitigate potential issues
- **Analysis of project-related information**, including site conditions, utility impacts, and regulatory requirements
- **Quality control, quality assurance, and quality verification** activities throughout the design and construction phases
- **Constructability reviews** to ensure the design is practical and efficient for implementation
- **Identification and acquisition of long-lead items** necessary for timely project delivery
- **Coordination and organization of subcontractor participation**, including bid packaging and scope alignment

The anticipated stipend payment to the Design Builders should also be added at this phase. Any stipends or payments used to offset the cost of preliminary design for responsive proposers must be included in the estimate.

The minimum stipend amount for each proposer is calculated in accordance with the WSDOT Construction Bulletin #2024-04, *Minimum Stipend Amounts for Design-Build Projects*. The stipend amount is paid to each proposer that submits a responsive proposal, and the stipend will be paid within 45 days after award of the contract or after the decision not to award the contract.

In accordance with the template Request for Qualifications (RFQ) document, WSDOT will short list three to five of the most highly qualified submitters to proceed to the proposal phase (this is when "submitters" become "proposers"). Accordingly, it would be best practice to multiply the calculated stipend amount by five for the estimated amount.

The stipend payment allows WSDOT to own approved Alternative Technical Concepts (ATCs) submitted with the proposals, and WSDOT could use them on the project. A proposer may refuse the stipend payment if they want to keep their ATCs proprietary.

Exhibit 12-2 Average WSDOT Expenditure of PE% for Design Build in CN

NOTE: These figures do not include the design builders share of CE%.

Average PE%	
Project Size	Sub Programs include – I1, I2, I3, I4
Final Engineers Estimate	Owner Concept Development PE% Range
> $2,000,000	13% - 19%

Data sources: CPMS vol. 100 and Financial Data marts as of 10/13/2021.

% shown is an average of the projects within the subprogram for Design Build projects.

Data extracted from project with operationally completion date prior to October 13, 2021

Cost range is based on final Engineer's Estimate.

12.04 Construction Engineering – Owner / Design Builder

The cost estimate should include two lines for Construction Engineering (CE) one for the Owner CE costs and one for the Design-Builder CE costs. Construction engineering is inclusive with quality control, quality assurance, and quality verification activities conducted during the design-builder's design and construction of the project. Construction engineering includes activities such as contract administration, design oversight, supervision and inspection of contract activities, measurement and payments, preparation of as-built plans, and final estimates.

Exhibit 12- illustrates the average WSDOT expenditure of construction engineering (CE) for design build projects by construction project size and sub-program.

Exhibit 12-3 Average WSDOT Expenditure of CE% for Design Build in CN

Note: This information offers perspective only. Each project team must consider specific attributes of their project and the CE amount estimated to appropriately fit the project context, environment, and requirements. These figures do not include the Design Builders' share of CE%.

Average CE %				
Project Size	Sub Program			
Final Engineers Estimate	I1 Mobility	I2 Safety	I3 Economic Initiative	I4 Environmental Retrofit
$0 - $500,000	No data	14%	No data	No data
$1,000,000 - $2,000,000	No data	8%	No data	No data
$2,000,000 - $5,000,000	14%	22%	No data	No data
$5,000,000 - $10,000,000	19%	14%	No data	12%
>$10,000,000	25%	8%	5%	9%

Data sources: CPMS vol. 100 and Financial Data Mart as of 10/13/2021.

% shown is an average of the projects within the subprogram for Design Build projects.

Data extracted from project with operationally completion date prior to October 13, 2021

Cost range is based on final Engineer's Estimate.

12.05 Contract Contingencies

In accordance with RCW 39.10.320 the project budget shall ***provide contingencies not less than five percent*** of the anticipated contract value. The estimate should include contingencies as a WSDOT cost during the construction phase, separate from WSDOT CE costs, and not included as part of the design-builder's cost estimate. Change order costs throughout the duration of the project should be fully covered within these contingency costs.

12.06 Right-of-Way

The cost of right-of-way includes the costs to research, and acquisition of right-of-way for the project, including construction or other project easements, storm water management, wetland mitigation, and other work outside the roadway prism.

This includes the contractual obligations with property owners to relocate fencing, reconstruct gates, and reconstruct road approaches, etc., if not included in the engineer's estimate. Consideration should be given to the cost of appraisals, negotiations, demo, property management, cost of GEC support or other consultant work.

Costs of any required relocation assistance and benefits for displaced individuals, families, businesses, governments, and nonprofit organizations, as well as the administration costs of all right-of-way activities must be included. If the extent of the right-of-way acquisition is not known, then a risk contingency should be added based upon historical settlements and awards for condemnation cases, which must include costs for attorneys, engineering research, witness research, survey, and staff time.

Also, the right-of-way acquisition schedule needs to be considered. Right-of-way acquisition costs can increase quickly in rapidly developing areas. Early acquisition of right-of-way based on environmental documents may save money and protect the right-of-way from development. Special acquisitions, such as those from government sites can be time consuming and costly.

Exhibit 12-3 illustrates the average WSDOT expenditure of preliminary engineering (PE) for Design Build projects by construction project size and sub-program. This information offers perspective only. Each project team must take into account specific attributes of their project and the PE amount estimated to appropriately fit the project context, environment and requirements during owner concept development.

12.07 External Third Party (e.g., Utilities and Railroad Adjustments)

Utilities have the potential to introduce significant risk to a project. Often compared to the Real Estate and Environmental elements as the highest for introducing risk. It is best to locate and avoid as many utilities as possible during the design phase and establish appropriate contingencies for utility adjustments. Project Offices should work closely with their Region Utility Office as much as possible to identify and manage utility conflicts.

Project Offices are highly encouraged to budget and plan for using subsurface utility engineering (example: a pre-qualified consultant) to help identify all utilities within the project limits and to inform utility adjustment strategies when needed. Early in the design development, your Region Utility Office can determine utility cost responsibility for all utilities impacted during a WSDOT project. WSDOT cannot legally pay for a utility relocation if the utility occupies the public right of way through a franchise or permit. This applies to WSDOT, county, and even city right of way. WSDOT is responsible for any costs associated with a utility's relocation if that utility has a documented property right (e.g. easement). A Region Utility Office can assist in determining cost responsibility and gathering easement or franchise documentation. Typical costs include construction, labor, equipment, materials, and even engineering design work. WSDOT must have an agreement with the utility owner in order to expend funds for this work. For more information on determining the need for and development of an agreement, consult the WSDOT Agreements Manual (available to WSDOT only).

12.08 Transportation Demand Management and Transportation System Management

The program cost estimate should include the costs for the development and implementation of the transportation demand management and transportation system management (TDM/TSM) strategies along the construction corridor and in the communities in the vicinity of the project during the construction period that are related to the project. These costs could be part CE Cost for the maintenance of traffic during construction and/or part of the final project as completed.

12.09 Risk and Uncertainty

Cost Risk Estimating Management is an integral element of project development that begins early in design. Project risks and uncertainties should be estimated using established guidance and best practices. All elements of project cost must be accounted and budgeted. Risk reserve is determined throughout the rigorous risk analysis process in accordance with Project Delivery Memo #25-01.

The Cost Risk Estimating Management process produces a project risk management plan. As the project is refined, the contingency should reflect the shift of these contingencies into actual cost categories. Any contingencies included in the cost estimate should be expressed in terms that can be easily presented to and understood by the public.

12.10 Allocation of Risk

The allocation of risk determines who is responsible for the risk. Risk should be allocated to the party that is best able to manage, control and minimize the impact of the risk.

WSDOT can manage risk in design build projects by judiciously structuring risk allocation for owning, transferring, or sharing risk within the RFP documents.

The appropriate balance must be struck as to not have an inappropriate burden of risk placed on the Design Builder. The outcome of unbalanced risk allocation is highly likely to be poor value for the Design Builder, resulting in a high-risk premium loaded into the price proposal for the project.

Risks typically allocated to the Design Builder include:

- Changes between preliminary and final design
- Design errors discovered during construction
- Final product quality
- Liability for defects either in design or construction

Risks typically allocated to the Owner include:

- Owner caused delays
- Obtaining access to the site
- Environmental permitting
- Litigation against the owner that stops the project
- Funding acquisition for the project
- Force majeure events (earthquakes, major flooding, rioting/war, unexpected discovery, significant changes in law, etc.).

There are some risks that may not have an obvious party for allocation and provide an opportunity for risk sharing. The allocation and distribution of these type of risks depends on the project.

Typical risks with variable and/or shared allocation:

- Site acquisition for temporary easements
- Changes in footprint of the project during design
- Negotiation or performance of utility relocations
- Changed site conditions
- Inflationary cost increases
- Any self-insured holdings or deductibles and for insured casualties

As an estimator, it is critical to understand how risk is distributed within the project framework. Risks may be allocated to either the owner or the design-builder, and this distribution can significantly influence how prospective bidders perceive and price the work.

Effective pricing should take into account:

- **The nature and magnitude of each risk**, including potential cost and schedule impacts
- **The party responsible for managing or mitigating the risk**, as defined in the contract or project plan
- **How each risk is documented and tracked** within the project's risk register, including any mitigation strategies or contingency allowances

A thoughtful and informed approach to risk-based pricing ensures that estimates are both realistic and defensible, aligning with market expectations and project objectives.

For additional information read WSDOT's Design Build Manual Chapter 3 on Risk Management. Allocation of risks defines ownership and responsibility for each task of the design build process. Though risk management should be a continuous process through the life of the project, there needs to be an especially strong emphasis during the initial design development and Request for Proposal (RFP) development phase of the project. Risk management should drive much of those processes. Risk is identified, assigned, and then mitigated through the development of both the project design and the RFP.

12.11 Value Engineering, VE

Although VE is not required for Design Build, many project managers have recognized and taken advantage of the benefits that a VE study can offer. VE can be used as an engagement and team building event during owner design and focus on design issues of concern. VE can also be used as a tool for improving the procurement process for the project, examples include validating delivery method selection; and/or engaging subject matter experts in particular areas of interest.

12.12 The Engineer's Estimate for a Design Build Project

Developing the Engineer's Estimate for Design Build requires an understanding of the project construction costs and design costs. The Engineer's Estimate should be updated with RFP addenda revisions during the procurement process. This may require adjusting the base estimate and rerunning the model in the CRA/CEVP prior to announcing the apparent best value proposer.

The Construction estimate includes the cost of the design-builder's materials, equipment, labor, overhead and profit, risks, and escalation. The construction estimate is typically done using unit bid analysis or other data to determine the construction costs for each element of the project.

The design cost considers two elements (line items), and both elements should be included in the construction phase of the program estimate:

1. The design-builder's CE costs
2. The design allowance costs - The remaining preliminary engineering and design that must be completed by the design-builder to bring the design from owner concept to Released for Construction (RFC) design. Use a % for design allowance as listed in the Exhibit 7-3, Markups Summary. The % should be chosen based on the appropriate level of design completed (i.e. if you would consider your conceptual design to be completed to a scoping level design, you would use a design allowance between 20% and 30% of the construction costs). Using a 30% design allowance is a good approach for design-build projects. This takes into consideration that the conceptual design is likely at a scoping level of design, and that the design-builder's design may significantly change from the conceptual design in the RFP.

The Engineer's Estimate is put together during owner concept and completed during the RFQ/RFP development. This means it is an estimate for a project that is in early design (say 20% to 30%). Hence the estimate accuracy and detail is commensurate with the level of design, and should be taken into consideration with your design allowance line item.

When performing a CRA or CEVP, the project team should discuss with the facilitator how to separate the design-builder's costs from the total construction costs.

The Engineer's Estimate should be within the cost range provided in the RFQ and ITP documents for the project. The Engineer's Estimate value (or in some cases, the Upset Amount) will be added to the spreadsheet that CAA uses to determine the apparent best value proposer, and will be compared to submitted price proposals.

After determining the risk loaded value to use for the Engineer's Estimate, there are two other items to consider during the procurement phase:

1. Breakdown of Bid items – Most of the project costs will be split into bid items 1 and 2 (there may be more if the project includes multiple locations, such as fish passage bundles), which are lump sum bid items, split between Tax Rule 170 and Tax Rule 171 respectively. Projects may contain other project specific bid items. The Engineer's Estimate is a risk based estimate, so the project team will need to incorporate the risk elements into the bid items. Typical bid items for a design-build contract can be found in the resources section, in the 'Contract Form-DBM Guidance' document, on the HQ Design-Build sharepoint site:

   https://wsdot.sharepoint.com/sites/eng/cn/hqconstr/dpb/SitePages/Resources.aspx
2. Prequalification requirements for submitters – The project team will need a breakdown of % cost by WSDOT work classification number. The % cost should express the ratio of the estimated cost in each work classification number to the total estimated design-builder construction cost. This information will be used to determine submitter prequalification requirements in the RFQ.

12.13 Bidding Climate Impact

Cost estimates or Risk Registers should consider the economics of major projects in the geographical area the project is located. For example, material manufacturers that would normally compete with one another may need to combine resources to meet the demand of the major project. Extremely large construction packages also have the potential to reduce the number of design builders that have the capacity or capability to do the work and may need to be split up into smaller contracts to attract additional competition. Also consider the ability of the Design Builders to bond the project. This should be a consideration in the CRA/CEVP Standard requirements are a bond for the full project costs..

Cost estimates should consider market conditions. If the economy is experiencing a downturn and there is more competition for projects, Design Builders will bid with less profit. Conversely, if the market is healthy and more projects are advertised, Design Builders will bid projects with higher markups. In addition, consider the timing of the bid solicitations. Industry Capacity

The number of potential qualified Design Builders that can bid on major projects are limited to those that have the capacity to construct the project. Design Builders who bid on major projects often bid on projects throughout the country. If other major projects are being advertised concurrently, it may have a limiting effect on competition and result in higher bids. If possible, rescheduling advertisement dates may be appropriate.

12.14 Price Adjustment in Contract Provisions

The price volatility of construction materials and supplies such as asphalt, fuel, cement, and steel can result in significant problems for Design Builders in preparing bids. In some cases, prospective bidders cannot obtain firm price quotes from material suppliers for the duration of the project. This leads to price speculation and inflated bid prices to protect against possible price increases. If price adjustment provisions are used in the contract to respond to this price volatility, a portion of the risk is transferred to the contracting agency, resulting in lower bids. However, since the contracting agency may have to increase its prices paid to the Design Builder, a reserve amount must be set aside and included in the overall cost estimate.

12.15 Highly Specialized Designs

Technology Cost estimates should consider the impact of any requirement to use first-of-a- kind technology, new materials, or methods of construction. This will likely be a consideration of risk as well as the base estimate.

12.16 Construction Time

The impacts of construction activities (e.g., sequencing, traffic control, haul routes, accessibility, geographic locations, roads damaged by construction equipment, and ponds that may be silted as part of construction) should be considered when developing cost estimates. Also, costs associated with rush hour restrictions and night work must be considered. For longer duration projects, there is a greater risk for impacts to the construction schedule. Construction scheduled in winter or rainy seasons should be accounted for appropriately, since there may be a higher risk in meeting construction schedules due to unforeseen weather delays. The same may be said when a major project consists of two phases by different contractors that are interdependent. Also, compressed, or accelerated construction schedules could potentially increase costs.

12.17 Construction Incentives

The cost for the Design Builder to meet quality/material and performance incentives must be included in the construction cost estimate.

12.18 Number of Concurrent Contracts and Contract

On projects where multiple construction contracts are underway at the same time, close coordination of construction activities and schedules may be required. The potential for one Design Builder to impact another contractor's activities is higher and may result in additional delays or coordination costs during construction. This cost may be considered as a risk in the CRA or CEVP.

12.19 Transportation Management Plans for Work Zones

Major projects often have complex construction traffic control and may have multiple construction contracts underway at the same time. The cost of implementing the Transportation Management Plan for work zones must be included in the estimate. Costs may also include incident management, public information and communication efforts, transit demand management and improvements to the local area network, which help improve safety and traffic flow through the project during construction.

12.20 Environmental Impacts

Major projects go through a thorough NEPA process. Due to the size and complexity of most major projects, there is often greater public and resource agency scrutiny during construction. This attention results in a greater likelihood that additional environmental mitigations may be required once construction begins. These additional costs should be considered when estimating preliminary engineering cost. This is from FHWA: https://www.fhwa.dot.gov/majorprojects/cost_estimating/guidance.cfm

Although the intent of a project may be to avoid environmentally sensitive resources, some degree of environmental consideration and analysis is required for all major projects. Any additional environmental avoidance, minimization, mitigation, remediation, and enhancement costs must be included in the cost estimate. Costs to mitigate impacts to natural resources, cultural resources, neighborhoods, etc., need to be included in the estimate. Resource surveys conducted as part of the NEPA process provide valuable information for refining these costs. Additionally, some major projects may have enhancement work that is not directly related to the project. This may include other transportation modes and non-transportation related work. These costs must be captured and included in the cost estimate.

A major project that has potentially significant social or environmental impacts or has opposition from environmental or community groups or regulatory agencies, tends to include more environmental mitigation which results in higher costs than those projects with relatively little impact or oppositions. Moreover, consideration of cost should be included for projects that include Intelligent Transportation System attributes, as well as need for Context Sensitive Design.

Chapter 13 Definitions

A comprehensive glossary for cost risk estimating management is posted at: Cost Risk Estimating Management (CREM) Glossary

13.01 Key Terms

Allowance – Additional resources included in an estimate to cover the cost of known but undefined requirements for an activity or work item. Allowance is a base cost item.

Base Cost Estimate – The term "base cost estimate" was developed by WSDOT for cost risk analysis and represents the reviewed and/or validated project cost estimate to be used in the quantitative risk analysis for a project. The base cost represents the cost that can reasonably be expected if the project materializes as planned, including PE, RW, and CN costs. The base cost estimate is unbiased and neutral—it is not optimistic, nor is it conservative. It does not include any risks, but does include the WSDOT standard construction contingency, since that amount is based upon historical usage. Base costs reported to program management shall be in current-year dollars (the un-inflated estimate). Refer to the Plans Preparation Manual 800.03(2).

Baseline – The approved time-phased plan (for a project, a work breakdown structure component, work package, or schedule activity), plus or minus approved project scope, cost, schedule, and technical changes. It generally refers to the current baseline, but may refer to the original or some other baseline. Usually used with a modifier (e.g., cost baseline, schedule baseline, performance measurement baseline, technical baseline).

Baseline Preliminary Engineering (PE) – The effort (budget/cost) of taking a project from planning through the scoping and design phases of project development. Planning and scoping typically have separate budgets but are encompassed under Design or Preliminary Engineering (PE). The terms "Design" or "Design Phase" are sometimes used interchangeably with PE.

Basis of Estimate (BOE) – Documentation to enable the agency to easily track changes to project scope, cost, and schedule. A well-documented basis of estimate and documentation of assumptions used can eliminate overlap of future estimate assumptions. This document provides a trail about what is known about the project. This allows project "knowns" as well as "unknowns" to be clearly identified. This documentation is important because multiple estimators may be involved on the project; complex projects in particular take years to develop and estimates must be completed multiple times.

Construction Contingency – A standardized markup applied to the construction cost of a project that accounts for uncertainties in quantities, unit costs, and minor risk events that typically take place during construction. Refer to the Plans Preparation Manual, Section 800.03, for guidance on estimating construction contingency.

Construction Engineering (CE) – The project management effort (budget/cost) applied to the contract execution and construction phases of the project. Refer to the Plans Preparation Manual, Section 800.03, for guidance on estimating the CE cost.

Contingency **-** An amount added to an estimate to allow for items, conditions, or events for which the state, occurrence, or effect is uncertain and that experience shows will likely result, in aggregate, in additional costs. Typically estimated using statistical analysis or judgment based on past asset or project experience.

Cost Estimate – A prediction of quantities, cost, and/or price of resources required by the scope of an asset investment option, activity, or project. As a prediction, an estimate must address risks and uncertainties. Estimates are used primarily as inputs for budgeting; cost or value analysis; decision making in business; asset and project planning; or for project cost and schedule control processes. Cost estimates are determined using experience and calculating and forecasting the future cost of resources, methods, and management within a scheduled time frame. (Source: Copyright 2007, AACE International, Inc., AACE International Recommended Practices, Number 10S-90)

Estimator – A knowledgeable professional assigned to evaluate the probable cost of projects.

Parametric Estimate – A method of estimating the cost of a project (or part of a project) based on one or more project-based cost factors. Historical bid data is commonly used to define parameters related to the cost of a typical transportation facility construction, such as cost per lane mile, cost per interchange or cost per square foot. Percentages can also be used to estimate the cost of project elements based on historical cost information.

Parametric methods are often used in early estimating, such as planning and scoping estimates. (Source: WSDOT working definition)

Risk-Based Estimate – An approach that involves simple or complex risk modeling based on inferred and probabilistic relationships among cost, schedule, and events related to the project. Risk-based estimating uses historical data and/or cost-based estimating techniques and an expert's best judgment to develop the project "base cost" (project cost if the project proceeds as planned). Risk elements (defined as opportunities or threats) are identified, assessed, and applied to the Base Cost and/or base schedule. Often, this is done through risk modeling to provide a probable range for both project cost and schedule —Probabilistic Risk-Based Estimating. (Source: WSDOT working definition)

Risk – The effect of uncertainty on objectives. (Source: ISO 31000, Risk management – Principles and Guidelines). Risks may be threats that potentially thwart project objectives, or opportunities that boost objectives.

Risk-based estimating is an approach that regards risks (opportunities or threats) as a discrete part of the whole project estimate, regardless of whatever combination of methods used to determine costs of composite items. Risks are accounted for in aggregate or, assessed individually as affecting cost and/or duration. Any risk contingency already inflating the cost or duration of existing base items is removed. Project risks are thoroughly identified and assessed, with the aid of subject matter experts as required, focusing on risks deemed most significant. Assessment considers the compound uncertainty of risk likelihood and risk impact combined. Valuation of these may be anecdotal (qualitative: "high", "low") or as a measure in standard units (quantitative: $ 3 million, 5 months, 1 in 4 chance). Quantitative valuation may then be input to a model, ideally one that reflects the interplay of cost and schedule, to provide a realistic range of probable project outcomes—Probabilistic Risk-Based Estimating.

Risk-Based Estimating – a methodology that systematically incorporates risks—both opportunities and threats—into project cost and schedule estimates. Instead of embedding risk contingencies within base estimate items, this approach identifies, assesses, and quantifies risks separately. It helps organizations make informed decisions about risk management and contingency allocation, ensuring that estimates reflect uncertainties rather than just historical assumptions.

Probabilistic Risk-Based Estimating – takes risk-based estimating a step further by applying probability distributions and statistical modeling—such as Monte Carlo simulations—to quantify uncertainties in cost and schedule. This approach considers both the likelihood and impact of risks, generating a range of possible project outcomes rather than a single deterministic estimate. By analyzing these probabilistic variations, organizations gain insights into the most probable cost and schedule scenarios, supporting more strategic decision-making.

Chapter 14 **Appendix**

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14.01 The Project Estimate

If we were to gauge importance by how frequently we are asked about any aspect of a project, Cost Estimating would be the TOP PRIORITY – Because no question is asked as often and repeatedly as “How Much does it cost?” & “Why”, throughout the project’s lifetime.

14.02 Why is the Estimate SO important?

Cost estimates determine how much money is needed over time to support key programs such as the Highway System Plan, Capital Improvement and Preservation Program, and the Biennial Program.

WSDOT does not have sufficient funds to design and construct every project that is needed, so we prioritize projects. Cost estimates can be used during benefit cost analyses to prioritize a list of projects within a program.

At some point in the project development timeline, this project cost estimate will become the baseline budget used to control the funds allocated to the project.

14.03 What Cost Estimates are used for:

- Obligate funds such as the biennial budget or construction at bid.
- By Program Management to determine when the funds are needed so when an estimate is combined with a schedule, a project cash flow or “aging curve” is prepared.
- Determining a fair price for contract work.
- Compare/analyze the contractor’s bids and whether to award or not.

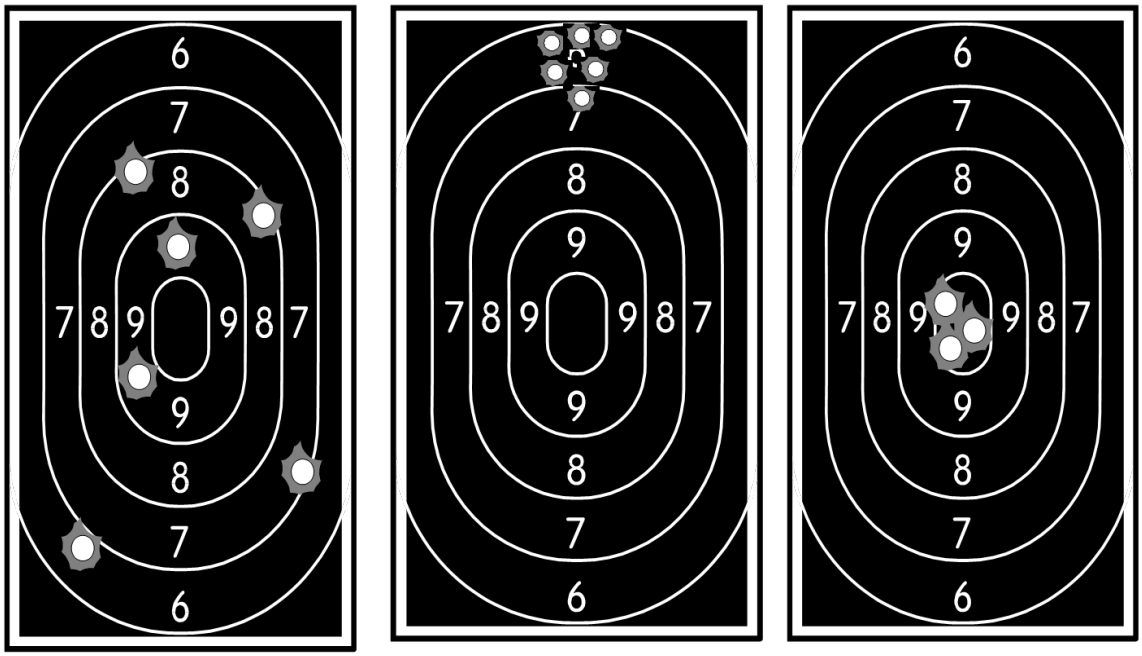
14.04 Estimate Accuracy.

Regardless of method used to prepare the estimate (bottoms up, parametric, bid history etc.) the estimate must be credible to be effective.

FHWA measures the departments’ ability to deliver “Effective & Credible” estimates. They state that to achieve consistently accurate estimates, or (*credible)* estimates, our estimates must be within +/- 10% of the Low Bid price at least 50% of the time on each of our projects.

The preparation and accuracy of the engineer's estimate should be reviewed if estimates are consistently higher or lower than the bids received, or if other anomalies consistently recur.

Exhibit 14-1 Estimate Accuracy



Neither Consistent nor Accurate	Consistent but not Accurate	Consistent and Accurate

What do you notice about the bullets in the figures? Even when they are consistant and accurate, they are not located one on top of the other.

In relation to cost estimates, this means that there is always some variation in cost estimate no matter how well defined the scope is for a project.

The challenge becomes, "How close can we get to the target or baseline cost estimate?

14.05 Creating and Maintaining Credible Estimates

Do not relegate estimating to a last minute after-thought. *Make estimation a priority* by allocating time and staff resources.



Integrate cost estimating into your project work-plans; it should be as well organized and planned as your other project development efforts.

Stay in shape (good estimating shape) – bid history is not enough. Consult with subject matter experts, industry and contractors, and other cost estimating resources.

Check Lessons-Learned and construction cost-trends; talk to others:

- Revisit bidtabs.net to analyze changing bid item cost trends for the major work elements, at least every 6 months to keep the estimate current.
- Make sure if the design changes**,** that the quantities in the estimate reflect the changes.

- As specs are developed, make sure they are considered as the estimate is updated.

A project baseline cost estimate is set during programming or early in preliminary design, endeavor to manage to this estimate throughout project development.
However, scope; requirements; market conditions and other variables continually evolve so, use tools such as Cost Containment Tables for timely decision making that indicate when projects deviate from the baseline.

Practice estimate transparency with disciplined communication of the project cost risks & uncertainties.

Estimators, at times, may be exposed to pressure from internal and external sources to provide low-cost estimates. Document your estimate cost decisions, record assumptions, costing sources, constraints, risks and any additional factors that affect the cost of your project. A well-documented estimate is a defendable estimate.

14.06 Cost estimation practice

1. ***Complete every step in the estimation process*** during all phases of project development. (use estimate quality checklists)
2. ***Document estimate basis,*** assumptions, and back-up calculations thoroughly. (create and maintain a "Basis of Estimate" document)
3. ***Identify project risks and uncertainties*** early and use these explicitly identified risks to establish appropriate contingencies.
4. ***Anticipate external cost influences*** and incorporate them into the estimate.
5. ***Perform estimate reviews*** to confirm that the estimate is accurate and fully reflects project scope.



14.07 FUNDING

Each project will have one or more sources of funding that are set up and assigned construction funds to pay for the appropriate work. Fund sources are created by combining the contributing participants, assigning their rate of participation and the sequence of participation (*who's money gets spent first*)

A Single fund may have as many as four participants. Each of these participants, contributing a portion of the money for the entire fund, that portion will be dependent on three criteria:

- Participation percentage
- Funding sequence
- Maximum Amount (an agreement for a contribution limit)

Participation Percentage is the participants "share" of the project cost. Some examples of this for Federal participation are:

- *Interstate Maintenance 90.66%*
- *STP Non-Interstate 86.50%*
- *NHS Safety 100%*

Participants MAY participate in more than one fund. All participants in the fund must be entered under the same fund number. Participants within the same fund cannot have the same funding sequence.

No fund may consist of only a single participant that has a Maximum Amount assigned. Maximum Amount participants must have a backup participant to pay for any costs that may occur over the maximum amount limit.

How Funding Sequence Effects Participant Contribution

Funding Sequence identifies what order they pay into the fund. This can have a significant effect on the amount of money they contribute.

EXAMPLE: In a $1,000,000 project, a participant who is the primary contributor (sequence 1) with a participation percentage of 50% will contribute $500,000. However, if they are the secondary contributor (sequence 2) placed after a participant who is participating at 80%, they will then only need to contribute $100,000. (*see an example of a Fund setup on the next page*)

When assigning Funding Sequence within a fund, the order of precedence is:

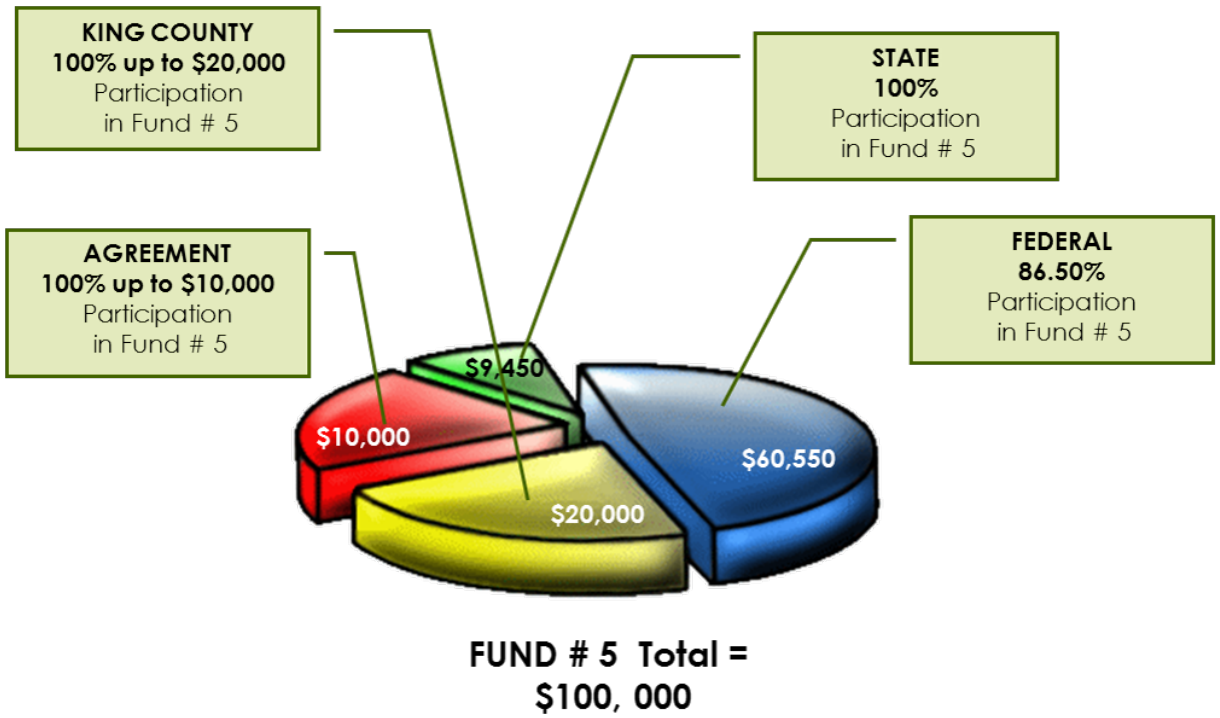
1. **Maximum Amount** (these entities have an agreed upon monetary limit to their contribution)
2. **Federal** dollars
3. **State** dollars - STATE dollars should always be placed as the final participant in any fund. Additionally, no fund should be created where there is only a single participant - other than STATE. *Exceptions to this may be arranged with Program Management.*

Communicate significant information about the fund that does not appear anywhere else in the estimate reports. It is an especially good idea to add descriptive fund specific text if the fund is an agreement but should also be used to describe types of Federal and State Dollars being used.

Exhibit 14-2 How Sequence and Percentage effect contribution

The Example **Fund 5** shown below includes four participants.

FUND 5				
In this Example All Bid Items Paid for by FUND 5 = $100,000				
Participants	**Fund Sequence (Order in which Dollars Contributed are spent)**	**Participation %**		**Actual $$$ Participant is obligated to contribute based on their Participation %**
King County	1	**100 %** With a Maximum contribution amount of $20, 000		$20,000
Agreement for $10,000	2	**100 %** With a Maximum contribution amount of $10, 000		$10, 000
Federal	3	**86.50%**		$60,550
State	4	**100%**		$9, 450



FUND # 5 Total = $100, 000

14.08 GROUPS

When is it required to create a group?

A separate group is required for the following:

1. Whenever there is a change in the program item number (PIN).
2. Whenever there is a change in program or sub program (I2, P1, P2, etc.).
3. Whenever there is a change in funding (any change in funding participants, their individual participation rates, or their source of funding).
4. Whenever there is a change in control section.
5. A separate group (one per project) is required for the bid item "Reimbursement for Third Party Damages".
    - This is the only item to appear in this Group.
    - This group must be a **State Funds** only group.

Separate Groups will also be required for work added by addendum.

Below is the Highway System Capital Program that a Group of work is programmed under. A contract may have multiple programs under which work has been programmed, but each EBASE Group may only have one. If your estimate contains work from multiple programs, a new group will have to be created for each one.

Exhibit 14-3 Capital Improvement Programs available for selection

Capital Improvement Programs available for selection			
D1	PROGRAM DEVELOPMENT	Q2	TRAFFIC OPERATIONS
D3	PLANT CONSTRUCTION	Q3	SPECIAL ADVANCE TECHNOLOGY PROJECTS
D4	PLANT MAINT. & OPERATION	R2	MAINTENANCE - OFF STATE SYSTEM
D5	OPERATIONS MGMT/SU	R3	SALES AND SERVICE TO OTHERS
E1	TRANSPORTATION EQUIPMENT FUND - STATE	T3	RESEARCH AND LIBRARY SERVICES
I1	MOBILITY IMPROVEMENTS	W1	TERMINAL CONSTRUCTION
I2	SAFETY IMPROVEMENTS	W3	EMERGENCY REPAIR PROGRAM
I3	ECONOMIC INITATIVES IMPROVEMENT	W4	WSF CONSTRUCTION (OBSOLETE 2-2007)
I4	ENVIRONMENTAL RETROFIT IMPROVEMENT	X1	OPERATIONS - VESSELS
I6	REGIONAL TRANSIT AUTHORITY	X2	OPERATIONS - TERMINALS
I7	TACOMA NARROWS BRIDGE	X3	OPERATIONS - MANAGEMENT & SUPPORT
I9	REIMBURSABLE INVESTMENTS ON STATE HIGHWAY SYSTEM	X4	MAINTENANCE - VESSELS
M2	MAINTENANCE FUNDS, ASPHALT	X5	MARINE MAINTENANCE
M3	MAINTENANCE FUNDS, SERVICES TO OTHERS	X6	MARINE MAINTENANCE - TERMINALS
M6	MAINTENANCE FUNDS, SAND/STOCKPILE	X7	MAINTENANCE - MANAGEMENT & SUPPORT
P1	ROADWAY PRESERVATION	Y5	RAIL CAPITAL - STATE
P2	STRUCTURES PRESERVATION	Y6	RAIL CAPITAL - STATE
P3	OTHER FACILITIES PRESERVATION	Z2	TRANS AID
P4	PROGRAM SUPPORT PRESERVATION	Z9	LOCAL PROGRAMS
P9	LOCAL INVESTMENT OFF STATE SYSTEM		

14.09 COLUMNS

Columns are used to separately identify and tabulate work within a Group.

Use Columns to create sub-groupings of the contract work, as an example mainline work should be distinguished from work on ramps, frontage roads etc.

GROUPS	**Columns → sub groups (of work)**		*Both of these Columns Will be assigned to the same Group for the work in this location. This will apply the financial data, tax rate, Program assignment etc. to the contract bid item work.*
GROUP 1 **I-5** **MP 99.00 to MP 104.56**	Column 1	Column 2	
	Mainline Paving	**Ramp Paving**	

Exhibit 14-4 Example of Column Setup

Each group is required to have at least one column associated with it.

Additional columns within a group are required for the following:

1. Each bridge and wall shall have their own column in order to identify the quantities of materials required at each of the structures.
2. Each state furnished pit site (mandatory and non-mandatory) shall have their own column.

Exceptions to rule 1 above:

- For projects with a single wall or a single bridge or both, the wall and bridge quantities may be entered into a single column or combined with another column.
- Paving across multiple bridges, these paving quantities need not be separated out for each bridge, they may be included in the mainline paving quantities.
- State Furnished Pit sites

Separate Columns will also be required for work added by addenda

Separate Columns may be used for work that may be cut from the project due to possible funding issues. These might include occurrences such as a Local Agency contributing to additional work as part of the contract, but this contribution is conditional upon their ability to secure the funding. By creating separate columns and associating the additional contract work bid items to these columns, it will make for quick revisions to the estimate by simply deleting those Columns of work, rather than time consuming re-calculation and distribution of Bid Items.

14.10 BID ITEMS

A bid item is a unit of work that is specified in the plans and detailed in the specifications. The work will be performed as part of the project by the contractor. Bid items are based on standard specifications, or special provisions that are included in contract documents to define work or procedures.

Standard Bid items are defined by a numeric code and have a specific unit of measure, section and prequalification code associated with them. Standard bid items represent work that is thoroughly

described in the Standard Specifications and well known to the contractor. Use these in your estimate when possible.

There will be work that is not used as often, unique or sometimes standard work that must be modified to fit the distinctive conditions of a project, this work will be represented by the use of non-standard bid items. These items of work must be fully defined by the user in the contract documents.

It is important that if using a modified "standard" item, where the work is being described or modified with a project specific special provision, that the standard item number is **NOT** used. The description must also be altered sufficiently so that the work is not confused with the standard item work that is otherwise described in the standard specifications.

14.11 Item Quantities & Distribution

The quantities for the bid item of work may be placed in a single location or several locations throughout the project. Where they are placed will affect the tax rates applied, the capital program it is associated with and who is paying for it.

A bid items' quantity may be distributed appropriately by adding columns that represent the location of work. The "Planned Quantity" of the bid item that will be placed, is then associated to the column or columns of work.

Additionally, when entering non-Standard items, the user will need to determine the appropriate sequence in the "Item Number" list to insert the item, and assign the Section Header, Unit of Measure and PreQual code for the item.

14.12 Non Bid or Below the Line Items

Non-Bid items are items of work or materials that are paid for as part of the total construction project, but are not performed by, nor furnished by the Contractor.

Some of this work may be partially funded with federal dollars or may be designated as entirely State funds depending on the work being performed. As such, they must be assigned to the appropriate Group and Coded as either a 700, 800 or 860 level items.

700 to 799 - number range have construction engineering and contingencies applied to them, but no sales tax. Federal Aid money may be used to fund all, part or none of this work. Examples of this type of work may include:

- Utility Agreements
- County/City Agreements
- State Force Work other than WSDOT i.e. WSP
- Engineering/Consultant Services

800 to 899 - number range do not have sales tax, construction, engineering, or contingencies applied to them. Examples of this type of work may include:

- State Force Work (WSDOT not to exceed $60k or $100k for Emergency work)
- State Supplied Materials (Parcel numbers, labor and payable agreements cannot be charged here)
- Some Permits and Agreements
- Rail Road Flagging (800 – 859 Range)

800 to 859 number range are to be used for Federal-Aid **participating** items. **860 to 899** number range are to be used for Federal-Aid **non**-participating items.

14.13 PARTICIPATING AND NON-PARTICIPATING ITEMS

Participating (*Eligible for Federal Aid*) - The following criteria must be met for charges to be considered participating when charged directly to a federal aid project:

- The work must be programmed with FHWA.
- The costs must have been incurred after the date of FHWA authorization to proceed with the work.
- The costs must be wholly for the benefit of the project being charged, e.g., labor performed directly for that project, supplies used up in the course of a project, etc.

Examples of federal aid participation -This list is not limiting or all-inclusive.

- Labor — Efforts to design, survey, prepare plans, appraise right-of-way, inspect construction activities, audit agreements, etc.
- Travel and Per Diem — of only those employees involved in any participating activity related to the project.
- Materials — incorporated into a federally approved project.
- Supplies — purchased and consumed entirely on a federally approved project.

Non-participating (*Not Eligible for Federal Aid*) - For specific projects, the following items are not eligible for federal aid participation:

- Equipment Purchase and Repair — unless specifically approved by FHWA, e.g., engineering, safety, or office equipment and supplies.
- Supplies — purchased and consumed on multiple projects or for the general management or operation of the organizational unit (example would be in maintenance or in administration — signs, small tools, and used temporary traffic control devices.
- Office Rental — including utilities and telephone service.
- Labor — for activities not directly related to the project nor programmed with FHWA.
- Labor activities pertaining to the general operation of an agency are not eligible for federal participation.
- Travel and Per Diem — of employees performing activities classified as non-participating or unrelated to a specific federal aid project.

NOTE: When Non-Participating Items are used, Accounting must be notified which funding source in the job will be paying for the work.

Per the Code of Federal Regulations 23 CFR 140 e Administrative Settlement Costs – Contract Claims participating Items are included in the following :

(a) Federal funds may participate in administrative settlement costs which are:

(1) Incurred after notice of claim,

(2) Properly supported,

(3) Directly allocable to a specific Federal-aid or Federal project,

(4) For employment of special counsel for review and defense of contract claims, when

(i) Recommended by the State Attorney General or State Highway Agency (SHA) legal counsel and

(ii) Approved in advance by the FHWA Division Administrator, with advice of FHWA Regional Counsel, and

(5) For travel and transportation expenses, if in accord with established policy and practices.

(b) No reimbursement shall be made if it is determined by FHWA that there was negligence or wrongdoing of any kind by SHA officials with respect to the claim.

14.14 ADDING AND DELETING ITEMS BY ADDENDUM

Prior to beginning revisions for addenda, be sure to create a new version.

NOTE: Once a project is turned in for Ad to the Region Plans Office, no additional changes may be made to the version that was submitted. Please coordinate with your Region Plans Office when creating a new version for Addendum revisions.

The following rules for adding and deleting bid items by addendum are to avoid changing bid item numbers (*sequence in the estimate listing*) for remaining items which would alter the contract documents as published for advertisement.

- All bid items added by addendum shall be added as the last bid items in the estimate.
- Items deleted by Addendum, must retain the bid item number in place, but change the change the Bid Item Status to "Deleted by Addendum. The quantity shall be revised to **.01** in column one, the unit price shall be **.01**. Only Column one should remain all other columns should be removed.

Exceptions to the rules:

Exception 1 - Items being revised for correction. The revised item can be added where an existing item is being replaced. In this case you can replace the old item with the new item.

Exception 2 - Alternate bid Items in your project.

When there are alternates in your project, when adding bid items to the **base** bid by addendum, add the items at the end of the **base bid item listing**. When adding bid items to **alternates** by addendum, add the items at the end of the **appropriate alternate** bid item listing.

Revisions made to the estimate after a project has been turned in for Ad that affect the bid items will require an addendum to be issued. These changes will need to be noted and identified on the Summary of Quantities plan sheets that will be included with the addendum.

14.15 The Reports

Proposal

The Proposal report is the Bid document used by the bidders to record and submit their bids.

Preliminary Estimate - Summary

This report shows the estimate summarized at project level and includes sales tax, engineering, contingencies, and non-bid items, as well as Contract Total. Additional pages of this report will be produced for each alternate combination.

Preliminary Estimate - Fund Report

This report shows the breakdown of fund participation summarized by federal-aid participating or non-participating and by fund name. Additional pages of this report will be produced for each alternate combination.

Preliminary Estimate - By Group

The Group report summarizes the estimate at the group level.

Preliminary Estimate - By Item

This report makes available each item along with its respective planned quantity, unit price and total estimated amount.

Work Classifications

This report groups the bid items into the prequalification work classes for the purpose of determining the predominate class of work in the project. Bid proposals are provided only to bidders qualified to perform at least 30% of the work in the project.

Summary of Quantities

The Summary of Quantities report is a working document where the estimating team can refine their column data, funding, group breakouts, items and quantities. This report is included in the plan sheets for each project.

Lump Sum Breakout Report

This report lists the breakout items that have been entered for each L.S. bid item where entries have been made for the items comprising the L.S. item.

Work Order Worksheet

This report is used by the Federal-Aid Program Engineer for reporting data to the FHWA.

14.16 Additional Resources for creating your estimate

- CPMS / Program Management
- Plans Prep Manual – Division 8
- Agreements
- TRIPS GEOMETRIC Report
- PROJECT SUMMARY
- Project Plans
- L.S Calc. Sheets
- Bid Tabs Pro / Unit Bid Analysis

14.17 Mobilization Percentage Guidance

Mobilization is a contract pay item that is used to cover a Contractor's preconstruction expenses and also the costs of preparatory work and operations. Since there is no defined list as to what this work effort is and each Contractor has the ability to adjust their bid as needed to cover these expenses, there are no hard and fast rules as what percentage should be used per Contract.

Exhibit 14-5 Mobilization Percentages

The following table is a suggested guide:

Contract Total		Percent of Mobilization
Less than $100,000	Use	8 % to 12%
$100,000 to $250,000	Use	6 % to 10 %
$250,000 to $500,000	Use	6 % to 9%
$500,000 to $1,000,000	Use	5 % to 9 %
$1,000,000 to $2,000,000	Use	6 % to 9 %
$2,000,000 to $5,000,000	Use	7 % to 9 %
$5,000,000 to $10,000,000	Use	8 % to 10 %
$10,000,000 to $20,000,000	Use	7 % to 11 %
Over $20,000,000	Use	7 % to 10 %

To more accurately calculate the Mobilization percentage that should be used for each individual project, consideration should also be given to:

- The location and complexity of the project
- The need for specialized equipment
- The type of work being performed
- Rural vs. Urban Projects with multiple work sites
- Excessive preparatory removal items
- Large quantities of excavation
- Spanning of multiple construction seasons when it would become necessary for the contractor to shut down and clear the work site between these seasons.

14.18 Project Percentages
14.19 (Change Order) Contingency

Contingency may be defined as "An amount added to an estimate to allow for items, conditions, or events for which the circumstances, occurrence, or effect is uncertain and that experience shows will likely result, in aggregate, in additional costs. Contingency is not used for:

- Major scope changes
- Extraordinary events such as natural disasters
- Risk reserves
- Escalation

Some of the items, conditions, or events for the effect is uncertain include estimating errors and omissions, minor price fluctuations (other than general escalation), developments and changes within

the scope, and variations in conditions. Contingency is generally included in most estimates and is expected to be expended.

All WSDOT design-bid-build projects are set up with a change order contingency percentage of 4%.

Please contact the Program Management office for guidance on using a Contingency percentage greater than 4%.

NOTE: for Design-Build projects, a contingency percentage of 5% is used.

For local agency projects administered by WSDOT off the State Highway system, no contingencies percentage will be set up.

14.20 Construction Engineering Allowance

Construction engineering allowance is determined as a percentage of the total contract item cost. This allowance is used to cover the cost to manage the overall planning, coordination, and control of a project from beginning to completion.

These percentages will vary by type of work and total dollar amount of the contract. On average, the department runs at around 15% engineering on all projects in the improvement and preservation programs.

Use the figures shown here to determine the historic-based Construction Engineering percentage for the type and size of project. Construction Engineering should be adjusted to reflect how a projects unique characteristics will affect the contract administration needs when determining the final percentage required to administer project construction.

The Region Program Development/Management staff can, based on appropriate justification, approve any changes in the construction engineering percentages for a project that differ from the rates listed. Copies of the approved justification letter shall be submitted with the final PS&E turn in for Advertisement.

14.21 Types of Preservation Projects

P1 Roadway:

(**PA**) Paving projects.

P2 Structures:

(**PB**) New construction, updating existing structure projects.

(**PC**) Seismic retrofits.

P3 Other Facilities:

(**PD**) Refurbish existing rest areas to extend service life and improve safety

(**PE**) Slope stabilization Projects

(**PF**) Construct weigh facilities

(**PH**) Major Refurbishments of electrical systems, electronics, mechanical systems and major drainage rehabilitation or replacement projects

Highway P – Preservation Program		
P1 – Roadway	**P2** – Structure	**P3** – Other Facility
PA - Paving / Safety Restoration	**PB** - Preservation	**PD** - Rest Area
	PC – Catastrophic Reduction	**PE** - Unstable Slope
		PF - Weigh Station
		PG - Program Support
		PH - Major Drain / Electrical

Exhibit 14-6 P- Program Project Percentage Table

Preservation Project Percentages								
	P1	P2		P3				
Project Cost Range	PA	PB	PC	PD	PE	PF	PG	PH
$0 - $250,000	24%	30%	36%	20%	30%	31%	20%	28%
$250,000 - $500,000	20%	30%	36%	18%	30%	24%	18%	28%
$500,000 - $1,000,000	20%	22%	32%	18%	25%	22%	15%	28%
$1,000,000 - $2,000,000	18%	20%	25%	15%	25%	22%	15%	20%
$2,000,000 - $5,000,000	15%	20%	22%	15%	18%	20%	10%	18%
$5,000,000 - $10,000,000	12%	15%	22%	15%	16%	20%	10%	18%
$10,000,000 +	12%	10%	20%	15%	10%	20%	10%	18%

14.22 Types of Improvement Projects

I1 Mobility — Improve mobility within congested highway corridors:

(**IA**) Congestion Relief Projects Urban

(**IB**) Congestion Relief Projects Rural

(**IC**) Bicycle projects

(**IQ**) High Occupancy Vehicle projects

I2 Safety — Provide the safest possible highways within available resources:

(**ID**) Accident reduction projects

(**IE**) Projects that improve roadway geometrics, eliminate at-grade intersections, install signals / channelization at intersections

I3 Economic Initiatives — Support efficient and reliable freight movement on state highways. Support tourism development and other Washington industries:

(**IF**) Freight and Goods improvement to all weather surfaces

(**IG**) Projects providing four-lane limited access facilities on a trunk system

(**IH**) Constructing Rest Areas

(**II**) Replacing or modifying structures on the Interstate System with, restricted vertical clearances and limited overload capacities

(**IJ**) Scenic Byway Projects

(**IR**) Bicycle rural road shoulder widening projects

I4 Environmental Retrofit — Retrofit state highway facilities as appropriate to reduce existing environmental impacts

(**IK**) Reconstruct storm water discharge facilities

(**IL**) Projects removing fish passage barriers

(**IM**) Projects including noise walls, berms, and noise mitigation measures

(**IN**) Projects for air quality

Exhibit 14-7 I - Program Project Percentage Table

Improvement Project Percentages												
	I1				I2		I3				I4	
Project Cost Range	IA	IB	IC	IQ	ID	IE	IF	IG	IH	IJ	IK	IL
$0 - $250,000	25%	20%	24%	25%	27%	28%	28%	22%	12%	12%	22%	35%
$250,000 - $500,000	25%	18%	24%	25%	28%	25%	25%	20%	12%	12%	22%	35%
$500,000 - $1,000,000	24%	18%	22%	22%	26%	19%	20%	18%	12%	12%	22%	32%
$1,000,000 - $2,000,000	22%	16%	22%	22%	23%	19%	20%	16%	12%	12%	20%	25%
$2,000,000 - $5,000,000	17%	16%	20%	20%	18%	19%	20%	14%	12%	12%	18%	20%
$5,000,000 - $10,000,000	17%	15%	18%	18%	14%	13%	14%	12%	12%	12%	18%	20%
$10,000,000 +	15%	15%	16%	12%	12%	10%	14%	10%	12%	12%	18%	20%

Exhibit 14-8 How to calculate Engineering Percentages for projects with multiple programs

